

Vedanta - Zinc Day

30 August, 2017



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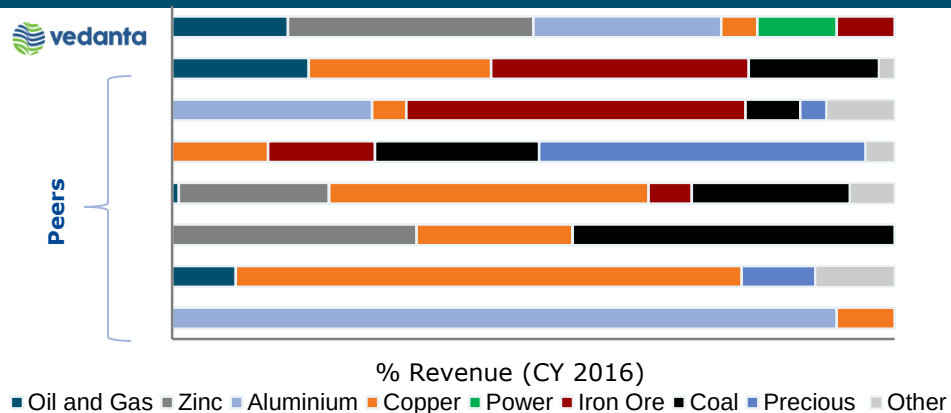
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Introduction

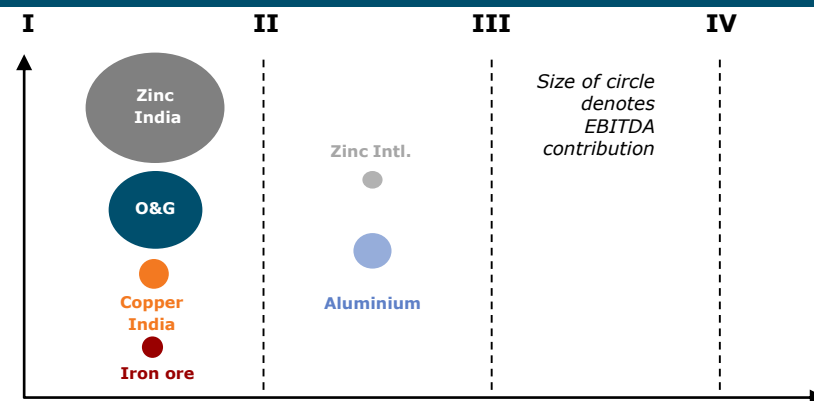
Arun Kumar G.R. –
Group CFO

Vedanta: A diversified resources company, with low cash cost positions, market leading growth and strong balance sheet

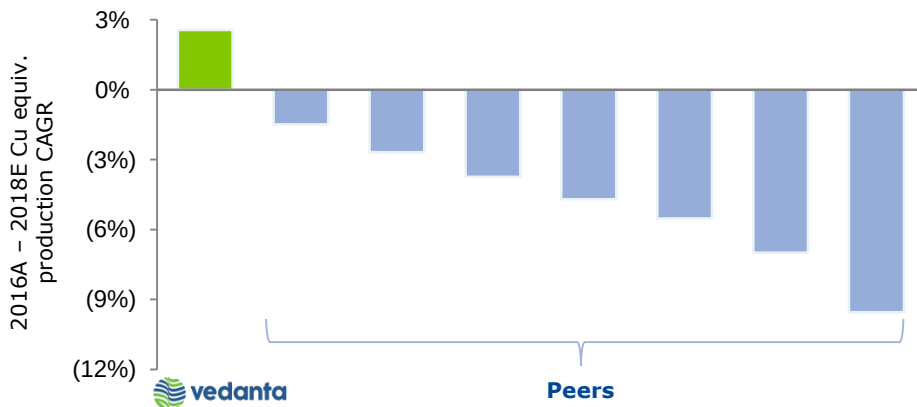
Commodity diversification¹



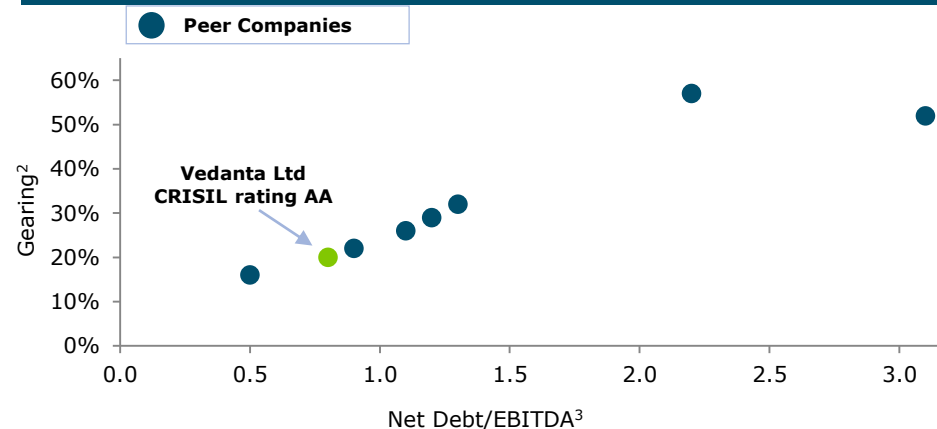
Attractive cost position



Sector leading growth



Strong balance sheet



Peers include BHP Billiton, Rio Tinto, Anglo American, Glencore, Teck Resources, Freeport and Hindalco

Source: Consensus, Company filings, Bloomberg, Wood Mackenzie, CRU for Aluminium; Company data for Vedanta

Notes: 1. All companies have been calendarised to a Dec YE; Glencore revenue split accounts only for their 'Industrial activities'; Revenues from copper smelting for Vedanta Ltd and Hindalco are based on benchmark Tc/Rc

2. Gearing is calculated as Net debt divided by the sum of Net debt and Equity (based on reported numbers)

3. EBITDA as per CY 2017 consensus estimates

A world-class natural resource powerhouse anchored in India

- 6th largest diversified resources company in the world¹
- Only global player with significant operations, expertise and majority sales in the Indian market – the fastest growing G-20 economy²
- Strong Balance sheet: Vedanta Ltd. Net Debt/EBITDA at 0.8x³ – amongst the lowest and strongest among Indian and global peers
- Completed Vedanta Ltd – Cairn India merger: Simplified group structure with greater financial flexibility to allocate capital efficiently
- Contributed INR 40,000 crores to Government of India in FY17
- 20% gender diverse leadership team; awarded one of the “Top Company to work for in Asia” by ACES, 2016

Notes: 1. As per CY2016 reported EBITDA

2. As per Moody's

3. LTM as of June 30, 2017

World class assets and operational excellence to deliver strong and sustainable cash flows

Production growth and asset optimization

Strong Shareholder Returns

- Announced dividend policy at Vedanta Ltd
 - pass through of HZL's regular dividend, plus
 - minimum 30% pay out of Attributable PAT (ex HZL PAT)
- HZL dividend policy - minimum 30% pay out

Maintain Strong Balance Sheet

- Continued reduction of gross debt
- Target for AA+ rating from current AA rating (CRISIL)

Grow Existing Businesses

- Focus on full capacity utilisation and production growth in existing businesses
- Any investment opportunities to clear hurdle rate of return



Production Growth and Asset optimisation



De-lever the Balance Sheet



Simplification of the Group structure



Protect and preserve our License to Operate



Identify next generation of Resources

Health, Safety, Environment & Sustainability

Phil Turner – Group Head,
HSE and Sustainability



HSE CHARTER

We will fulfill our duty to protect the safety and welfare of our people and plan to achieve our health, safety and environment goal of zero harm to become the best performer in the resources industry by:

- ✓ Implementing world class standards that deliver our legal obligations and company policies
- ✓ Managing risk in all forms by applying controls and testing their effectiveness
- ✓ Setting priorities and commitments and measuring, monitoring and reporting performance
- ✓ Maintaining an open dialogue with our business partners and stakeholders
- ✓ Applying this discipline every day in every aspect of our business

It's all of our responsibility to ensure that everyone who works on our site returns home safely every day while protecting the environment for our future generations.



Anil Agarwal
Chairman

June 2016

Our Vision

Zero Harm

Zero Waste

Zero Discharge

Deliver world class performance by identifying and managing risk and by benchmarking best performance

Engage with stakeholders and deliver best social responsibility performance in the resources sector

Eliminate fatal accidents

Ensure our people are not exposed to harmful levels of toxic substances

Deliver best in class performance in air, land, water & waste

Manage carbon reduction and GHG emissions in line with Indian and International norms



SZ safety intervention



BMM 1 000 000 fatality free shifts



Paper recycle campaign

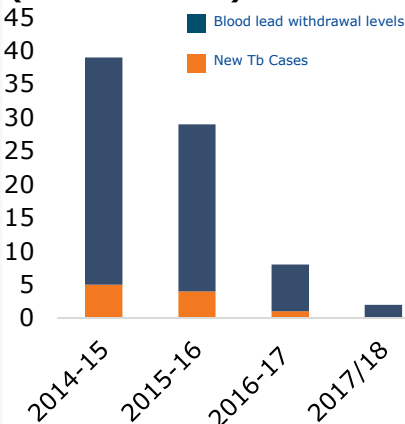


Voluntary HIV/Aids Testing

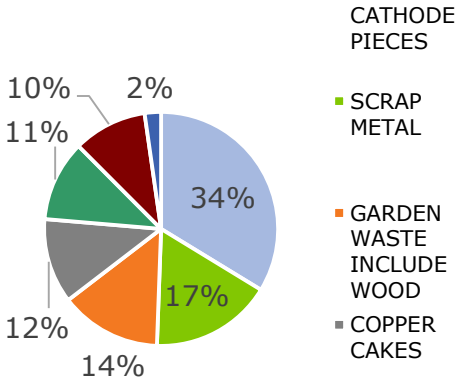
‘ZERO HARM, ZERO WASTE, ZERO DISCHARGE’ to all people and environment

- To be a caring organisation for our employees and **all stakeholders**
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- Focus on elimination of occupational health diseases
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- We aim for 100% compliance to Internal performance standards and processes

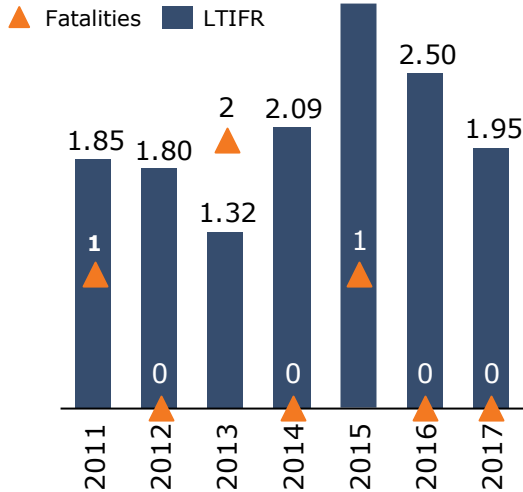
Occupational Exposure (# of incidents)



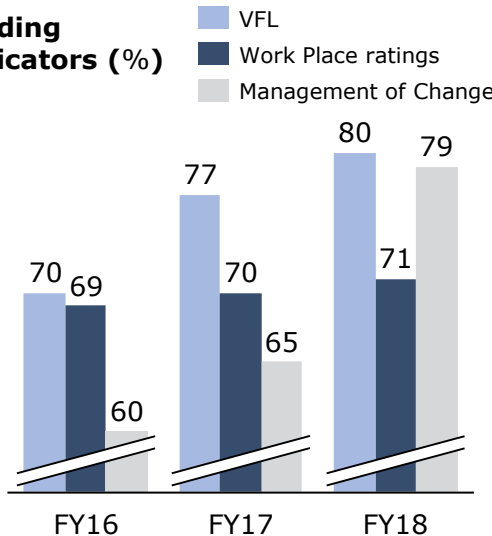
WASTE TO VALUE

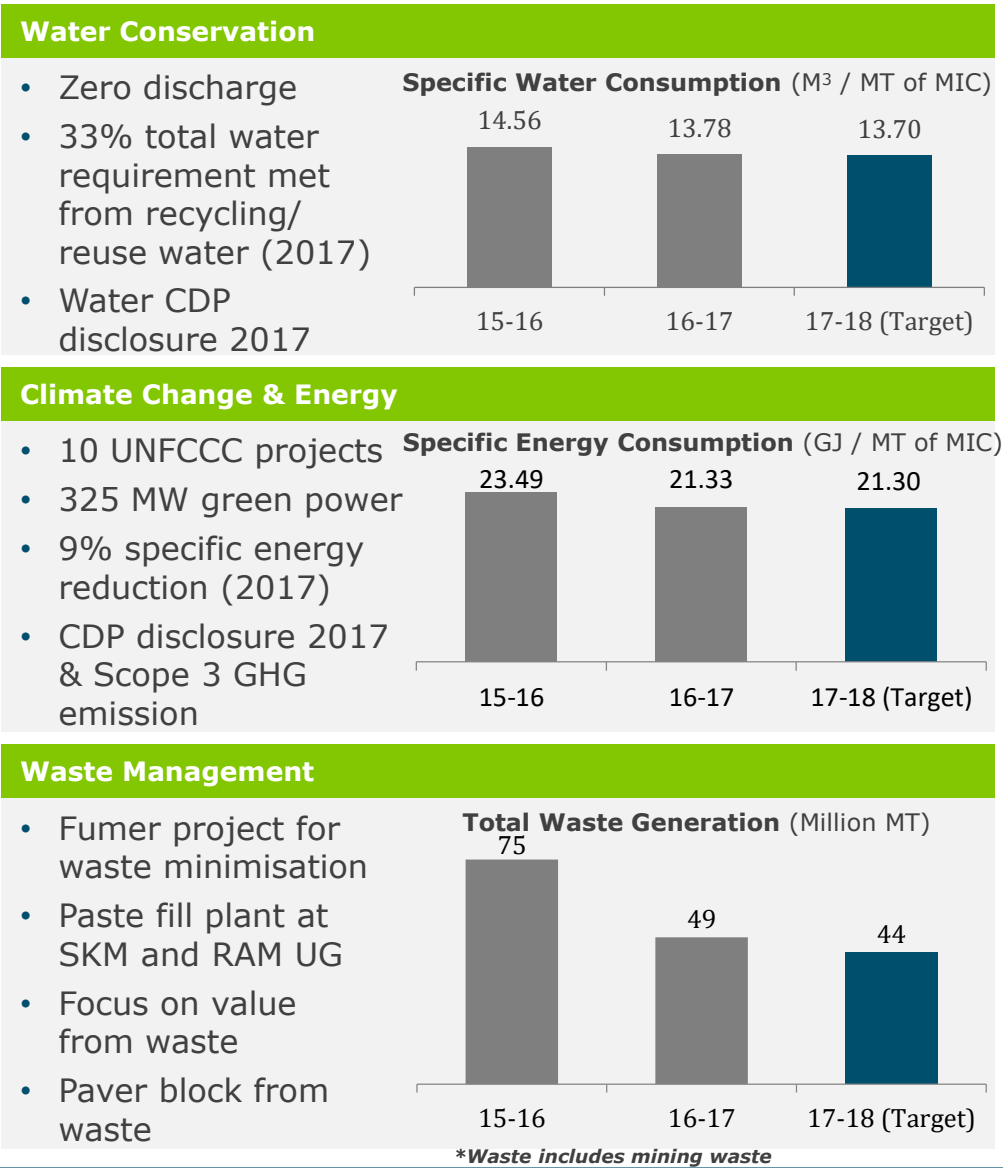
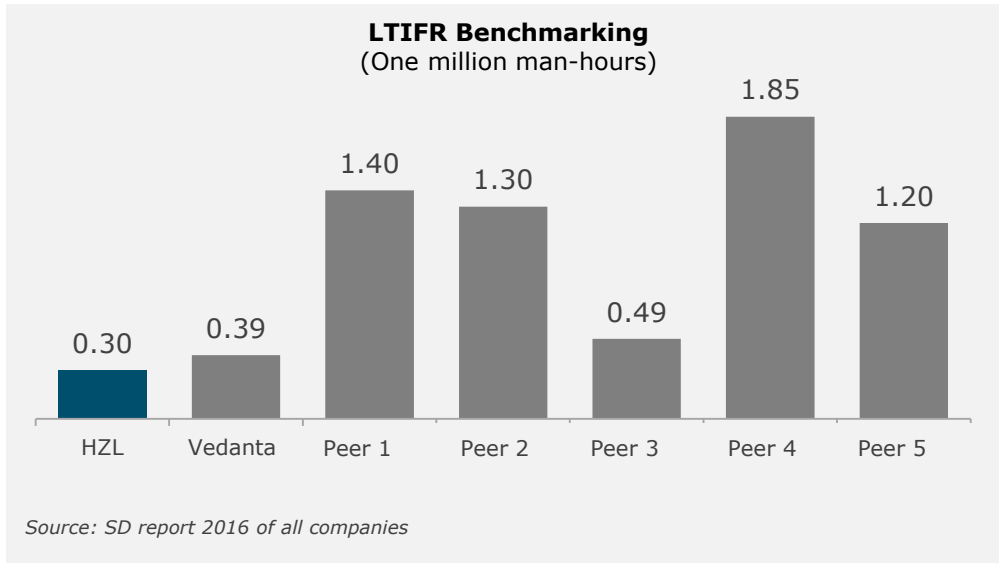
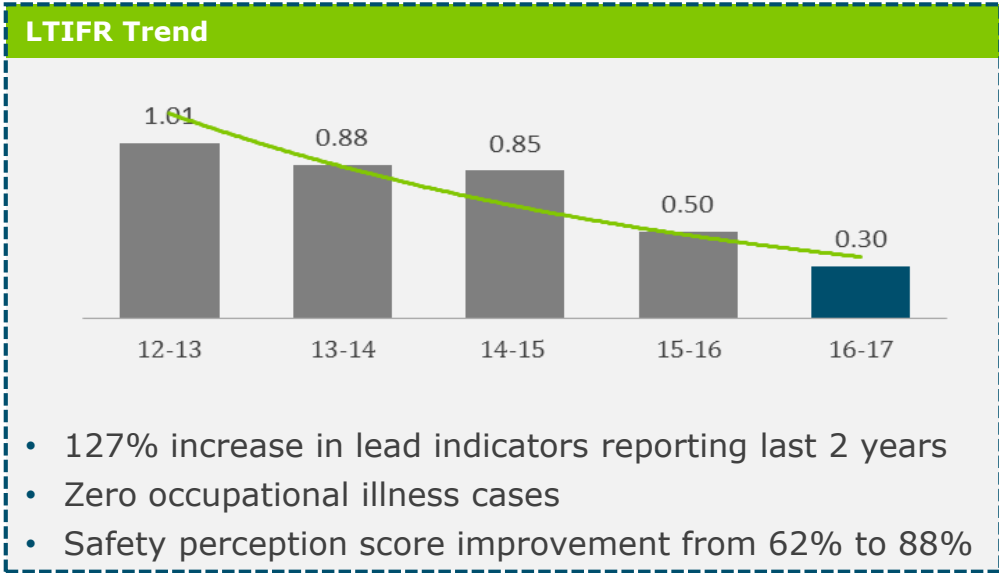


LTIFR (per million man hours worked) vs. Fatalities



Leading Indicators (%)



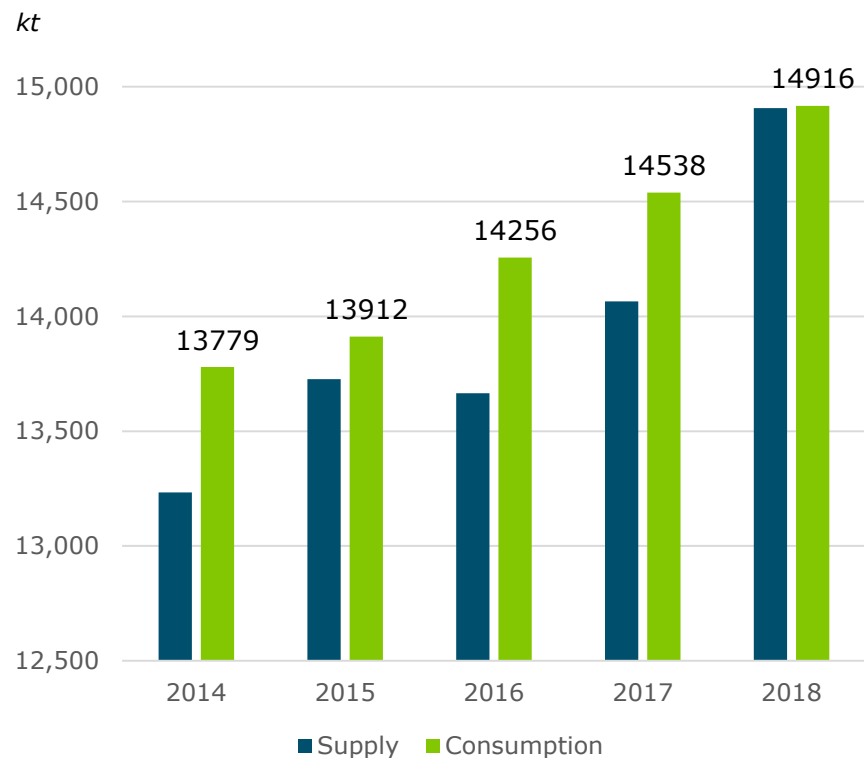


Zinc Market Overview

Sunil Duggal – CEO,
Hindustan Zinc

Global Zinc Market Overview

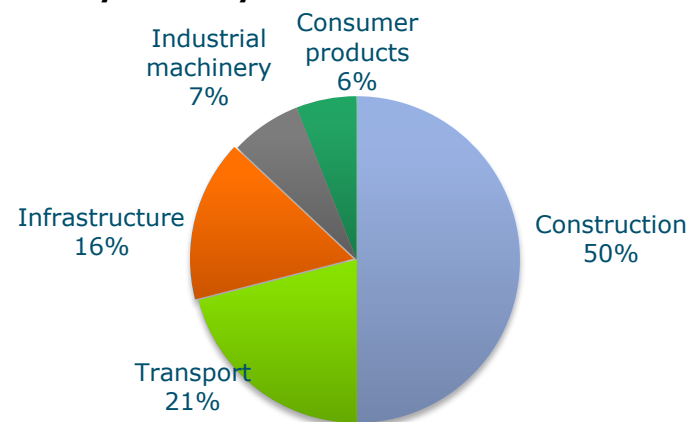
Refined Market Balance



Source: Wood Mackenzie LTO Q2 2017

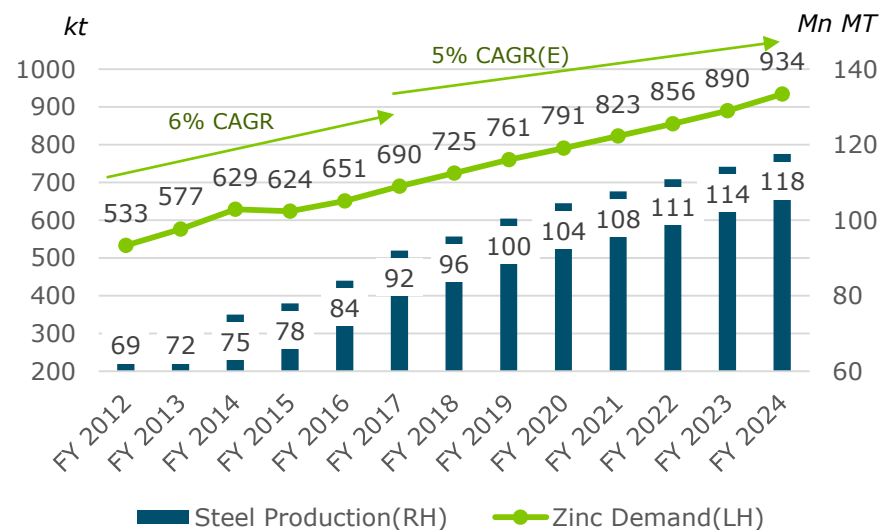
World - Steady demand growth at 2.6% in 2018
China - CAGR (2017-2020) of Consumption at 3%
India - CAGR (2017-2020) of Consumption at 5.1%

Zinc Demand By Industry



Source: Wood Mackenzie

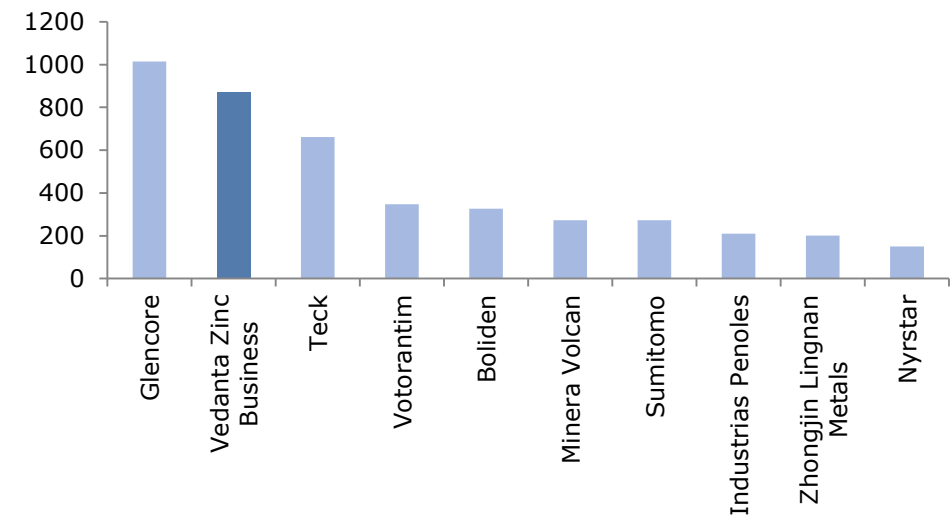
India Steel Production and Zinc Demand



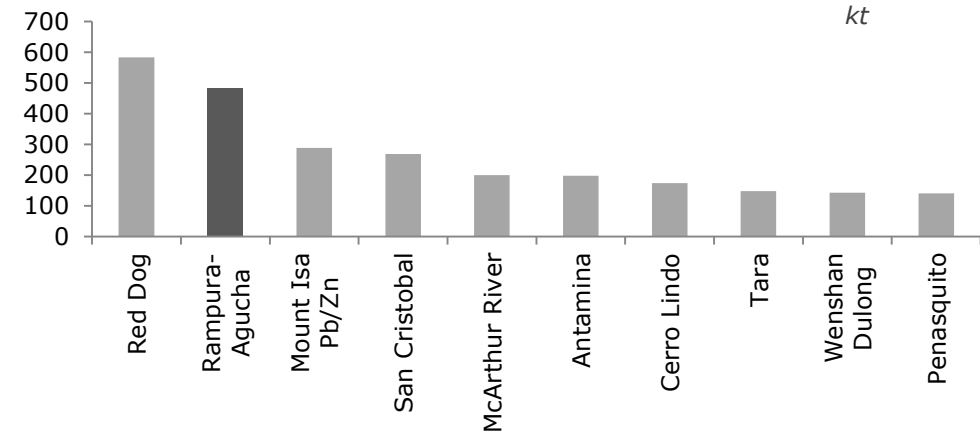
Source: Feedback Consulting, HZL and Wood Mackenzie LTO Q2 2017

MINES - 2016 *kt*

By Company

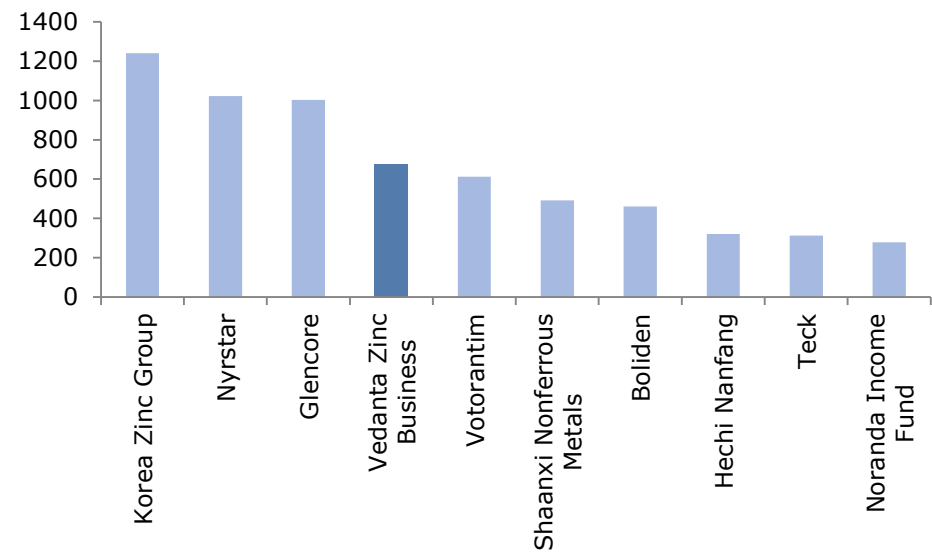


By Mines

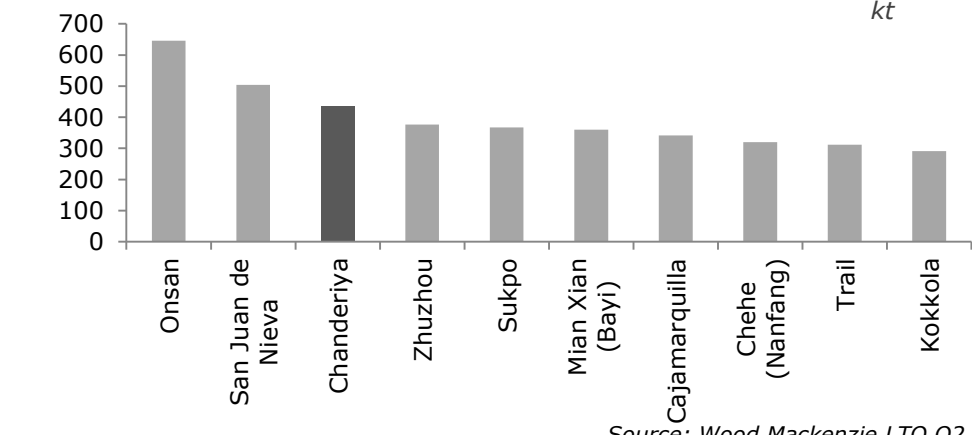


SMELTERS - 2016 *kt*

By Company



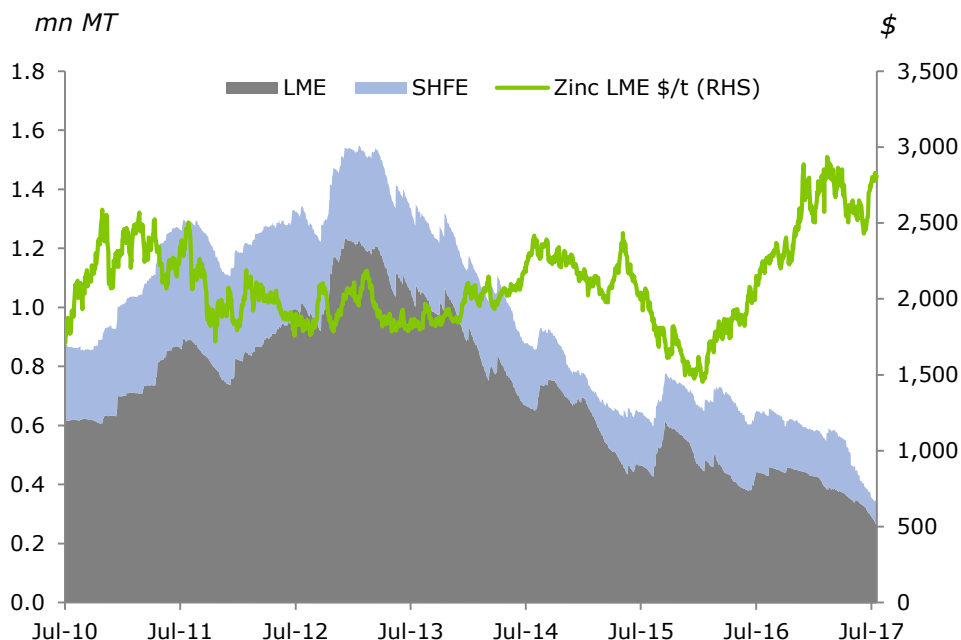
By Smelters



Source: Wood Mackenzie LTO Q2

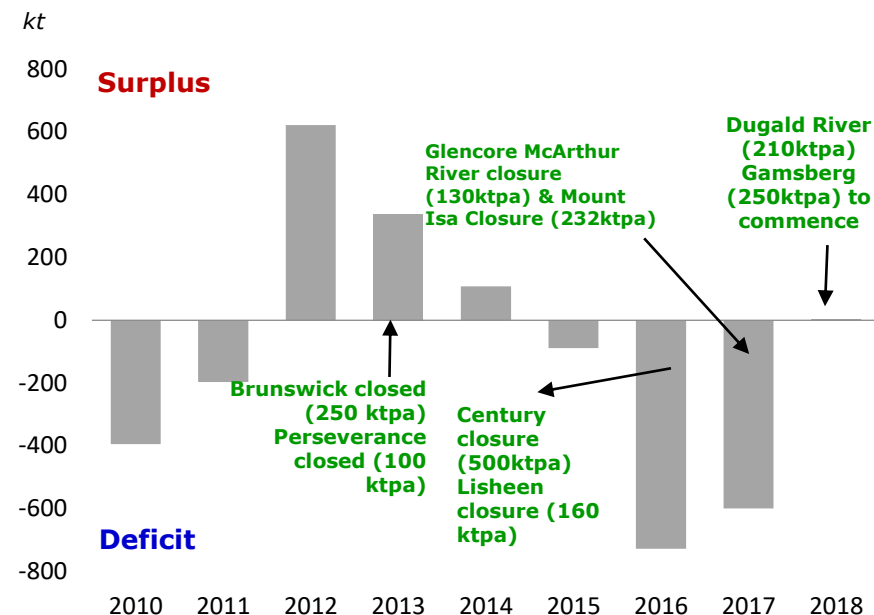
Strong Fundamentals

Declining stocks at LME warehouses



- Exchange stocks declining: 11 days of global consumption in July 2017; seven year low
- Zinc prices strengthening on continued supply deficits and declining inventories

Global Zinc concentrate deficit supporting zinc prices



Source: Wood Mackenzie LTO Q2 2017

- Gamsberg project to start in 2018
- 2018 growth assumed from resumption of Mt Isa & McArthur River as also Dugald River project
- Chinese mine capability to grow by 0.7 mtpa by 2021, with a further 1 mtpa from new projects

Source: Wood Mackenzie LTO Q2 (Conc.)

Vedanta - Zinc Day

Zinc International

30 August, 2017



01

Health, Safety & Environment -

Vision to Zero

Deshnee Naidoo - CEO

02

Sustainability – We take people along

Nora Ndopu – Head of CSR

03

About us –

Significant footprint with strong track record

Deshnee Naidoo - CEO

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Exploration Potential –

Unrivalled Zn address in Africa

Markus Schaefer – Head of Exploration

05

Gamsberg – Anchor for Growth

Satish Kumar –VP Projects

Liesel Jacobs – Project Manager

06

BMM – Enabling a world class Zn complex

Andre Trytsman – GM

07

Skorpion – Surpassing LOM expectations

Irvinne Simataa –GM

08

Digitalization & Technology -

Driving efficient growth

Dave Payne - Head of D&T

09

Outlook –

The path forward

Pushpender Singla - CFO

Q&A

01

Health, Safety, Environment- Vision to Zero

Deshnee Naidoo - CEO



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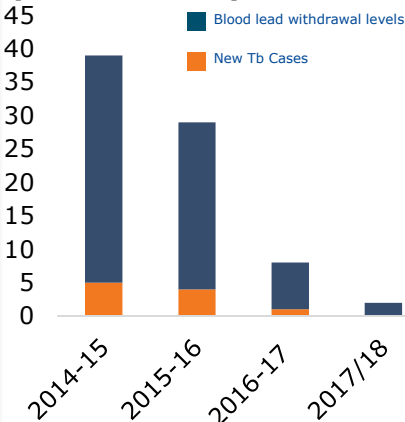


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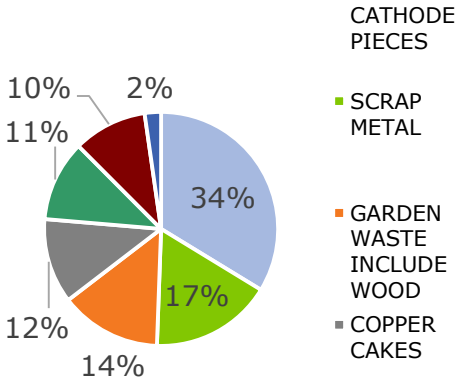
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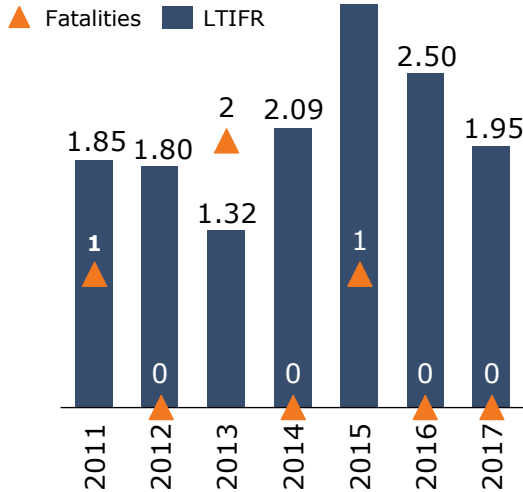
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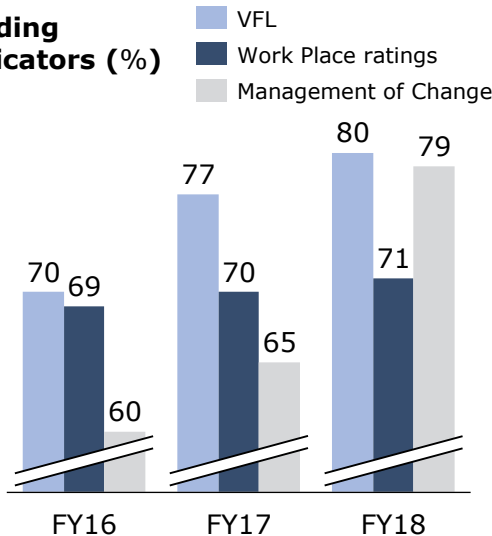
WASTE TO VALUE



LTIFR (per million man hours worked) vs. Fatalities



Leading Indicators (%)



02

Sustainability - We take people along

Nora Ndopu – Head of CSR

CSR
Last 5 years spend
>ZAR 60 million

- Livelihood
- Sports
- Health
- Education



Biodiversity
No net loss

- Bird Pond at SZ
- Rehabilitation Nurseries



People
>4000 employees

- Dedicated Teams
- Community Consultations and awareness



Social Licence to Operate

- Strive for beyond regulatory compliance
- Multi-prong CSR strategy focusing on health, education, sports and livelihood projects
- Preferentially target the empowerment of affected mining communities (women specifically)
- Supported by key stakeholder engagement
- Local employment (Direct labour: 1,497, Indirect labour: 2,675)
- Impacting approximately 40,000 beneficiaries through CSR

03

About us –
Significant
footprint with
strong track
record

Deshnee Naidoo - CEO

ZI's vision is to be the safest, socially responsible 1 Mtpa integrated Zn producer in the 1st quartile of the global cost curve

Skorpion Zinc

- Open pit zinc mine and refinery
- Largest integrated zinc operation in Africa
- R&R of ~26 MT with 2.7 MT of contained metal at average grade of around 9%)
- LoM of 4 years
- Workforce of 1,500
- Refinery capacity of 150 KTPA Special High Grade (SHG) zinc



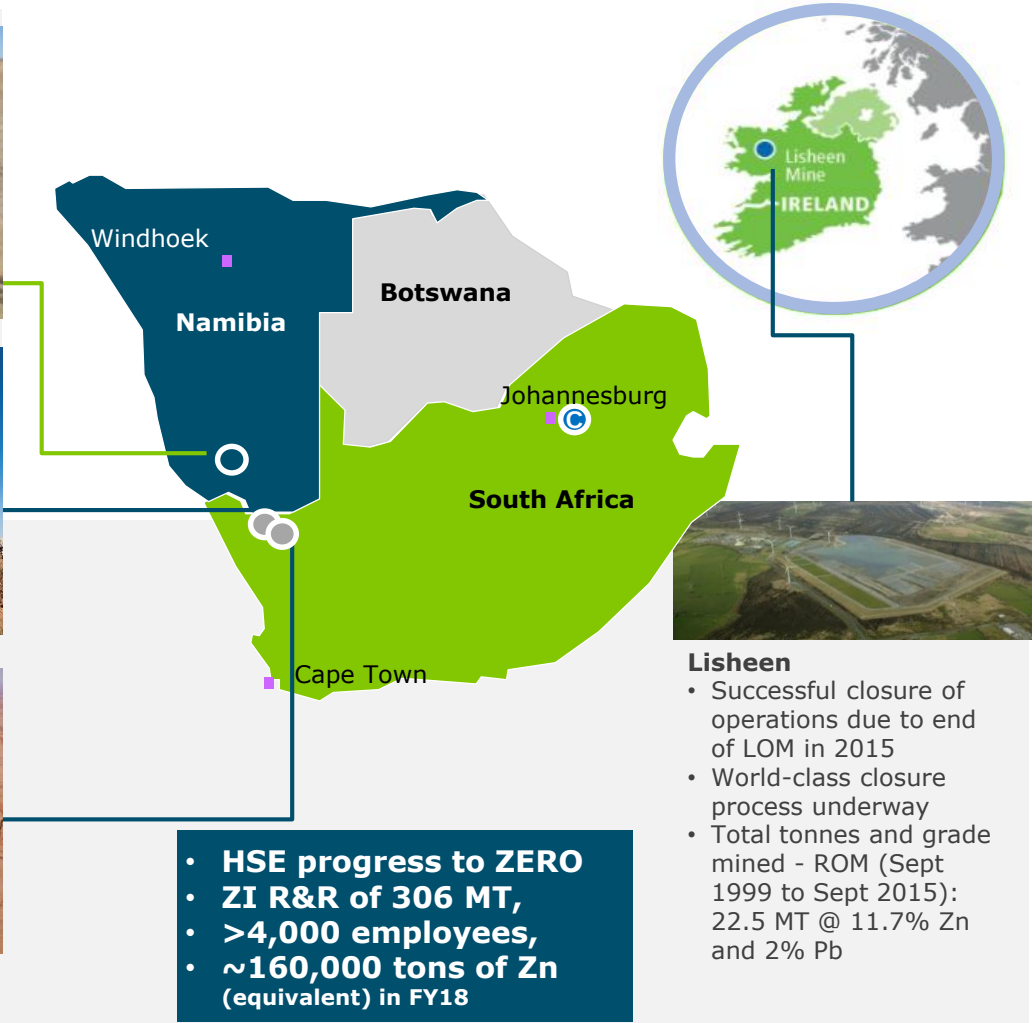
Black Mountain Mine

- An underground operation, mining zinc, lead, silver and copper deposit
- R&R of ~65 MT with 3.1 MT of contained metal a grade of around 5%)
- LoM of 4-5 years
- Workforce of 1,350 (80% of the employees are from the local region)



Gamsberg Project

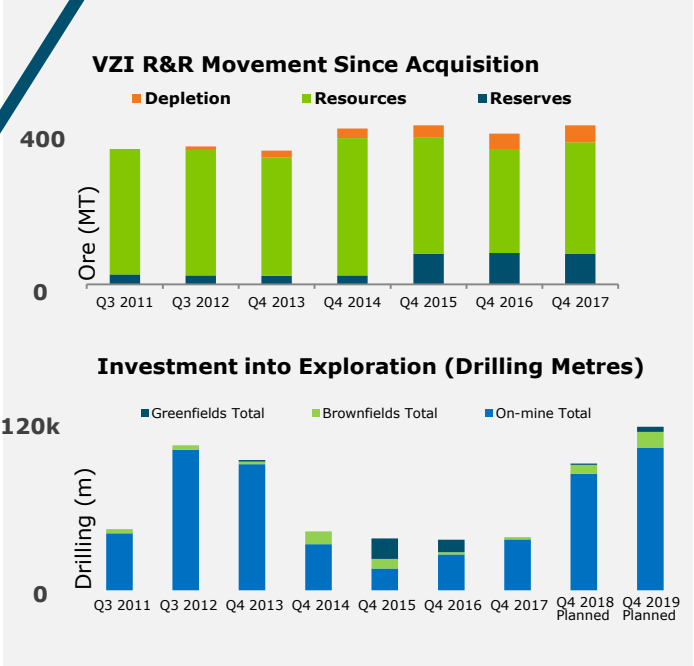
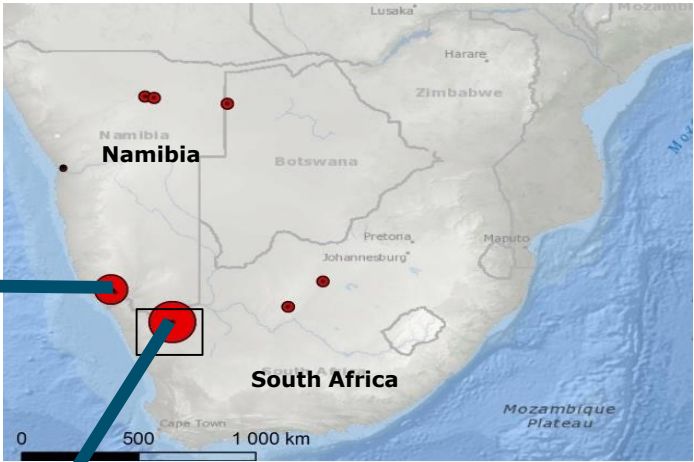
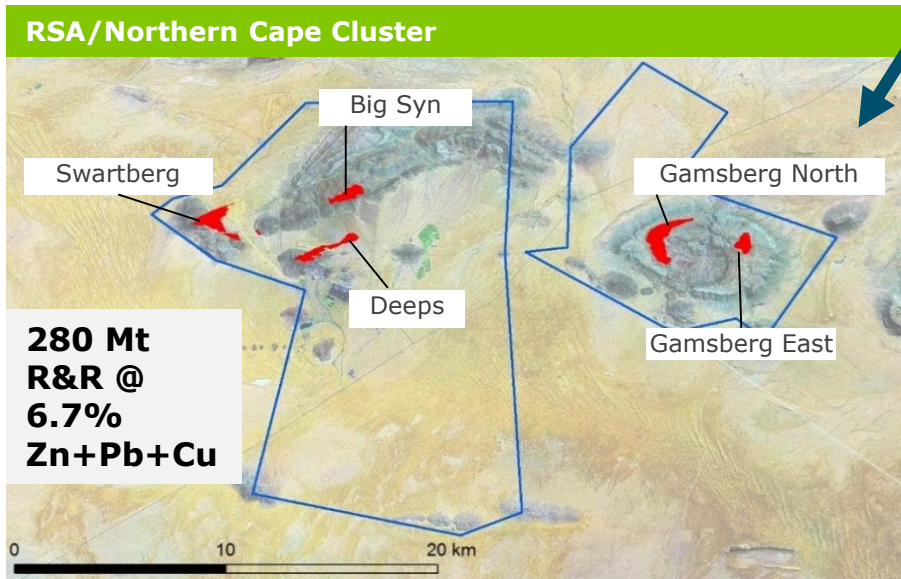
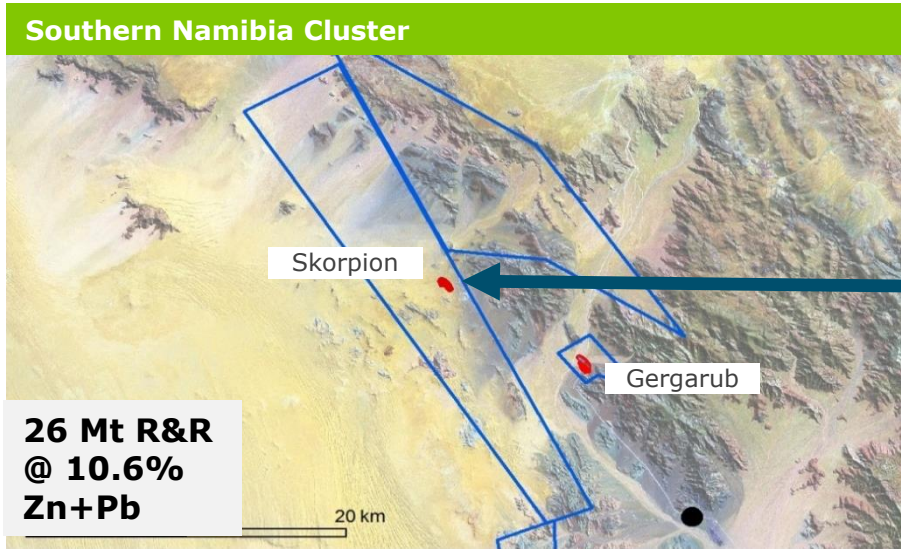
- One of the world's largest known Zn deposits
- R&R of ~215Mt with contained metal of 15.8Mt
- LoM of +30 years
- Phase 1 of project in execution, 400 mil USD capex investment, 250 Ktpa metal in concentrate.
- First production by mid CY 2018.



04

Exploration potential – Unrivalled Zn address in Africa

Markus Schaefer – Head of
Exploration



High value Resource Base to enable long LOM and expansions

- Total of 21.5Mt of Metal in R&R

Best Zn Address in Africa

- Extensive tenement portfolio (>5,000km²)
- Opportunity to double R&R

Aggressive Exploration Programme

- R&R including depletion has increased by 11% since acquisition to 24.4Mt metal
- Exploration throughout downturn. Significant step-up in 2017
- Use of state of the art technology (geophysics) – high discovery rate

05

Gamsberg - Anchor for Growth

Satish Kumar – VP Projects
Liesel Jacobs – Project
Manager



Gamsberg – Anchor for Zinc International Growth

Total Potential for Gamsberg 250 KTPA to 450 KTPA to 600 KTPA

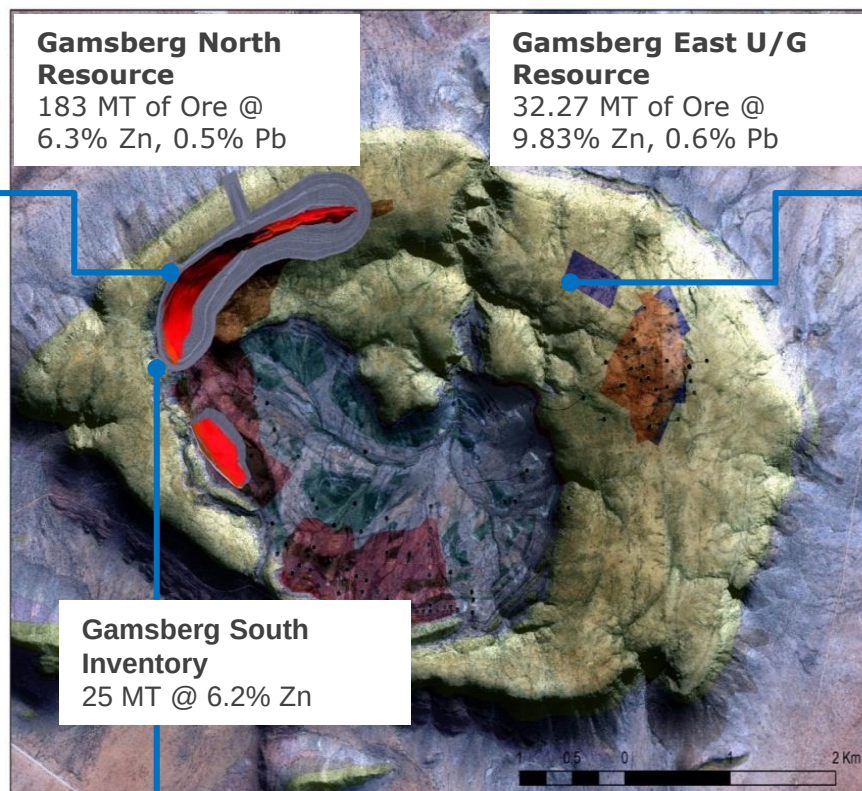
Phase 1 Project In Execution

Gamsberg Project (Phase 1)

- Current project (open-pit mine, concentrator and related infrastructure) 4 MTPA ROM : **250 KTPA MIC**
- **US\$ 400 mn** investment,
- 4 MTPA ore, 250 KTPA MIC
- **First production by Mid CY 2018**, 6-9 months of ramp up to full capacity.
- COP at US\$ 1,000-1,150/ton, First quartile of cost curve.
- ~ 1,500 jobs during construction, 850-900 jobs once operational

Gamsberg Mega Pit (Phase 2)

- Expanded open pit by 4 MTPA ROM
- 2nd concentrator stream of 4 MTPA
- Additional metal production - **200 KTPA MIC**



Gamsberg U/G (Phase 3)

- New underground option for Gamsberg East 2.5 MTPA
- Potential Expansion of Gamsberg North open pit to U/G
- +150 KTPA MIC

Gamsberg Extension areas (Beyond Phase 3)

- Good Potential
- Extensive drilling required to firm up potential

Gamsberg Phase 1 – Project Scope & Capital Optimisation

Gamsberg Open-Pit layout

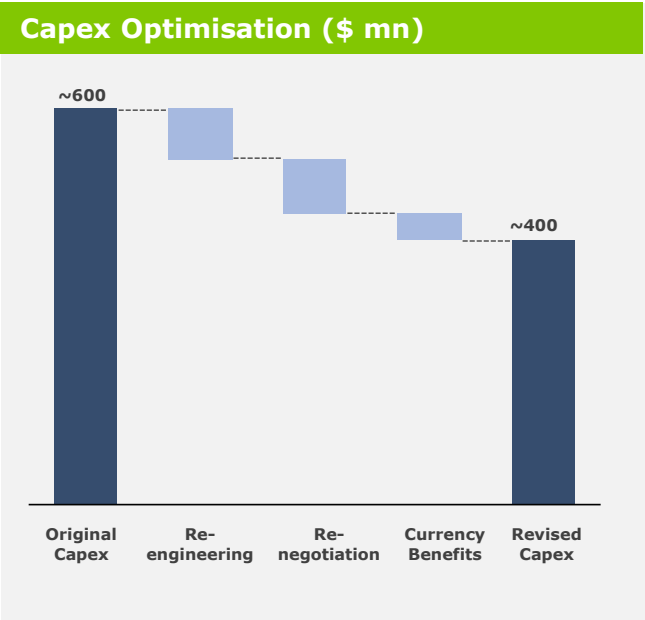
- Open pit – Drill and blast, load and haul
- Pre-stripping 65 -70 MT, Ore production @ **4 MTPA**
- Total mining @ **35 MTPA** rate

Gamsberg Concentrator

- **4 MTPA** Ore feed capacity
- **250 KTPA MIC Production**
- Crushing, milling, flotation, dewatering, utilities and tailing pipe-line to tailing storage facility

Gamsberg Tailings

- Construct tailings dam to store **3.55 MTPA** of tails
- Located 3.2 km north of the plant
- Dam will have HDPE liner



Support Central Facilities

Power

- **Power: 40 MVA** transmission line from ESKOM sub-station (20 km), 66/11 KV MRSS at Gamsberg

Water Package

- **Water:** New pipeline from Orange River (38 km) **13 MLD**

Building Infrastructure

- **Building & Infra:** Offices and building infrastructure, para-medics, etc
- **Housing: 350** housing units in phases

Biodiversity

- 85,000 plants relocation
- Acquire habitat (12,900 Ha) as per Biodiversity Offset agreement

- **Original Gamsberg capex was reduced from ~\$600 mn to ~\$400 mn by:**
 - Value engineering
 - Re-negotiation
 - Exchange rate benefits
- **COP improvements of ~10%**
- **Improved project financials**

Gamsberg Phase 1 – Project Execution Philosophy

Gamsberg – the new way of delivering projects in Vedanta

- An outsourced approach
- Leveraging owners engineers and external skills
- Lean project structure
- Minimum number of packages
- High governance standards
- Latest technology & digitalisation initiatives

1

Outsourced Model

- Mining - Fully outsourced including operations and infrastructure
- Mining - Variable cost model based on output
- Plant & Infrastructure – Fully turnkey with least number of packages
- Plant & Infrastructure - Complete system performance/process guarantees

2

Capex Optimisation

- Capex reduced from US\$ 600 mn to US\$ 400 mn
- Collaboration with EPCs and equipment vendors to reduce cost
- Value/re-engineering by EPCs and owner's engineer

3

Organization Structure

- Lean owner organisation structure
- Project technical specialists outsourced with project objectives KPIs
- Engineering and construction management support from owner's engineer

4

Governance

- All licenses and permits in place
- Packaging philosophy - minimum number of orders
- Contract quality and contract management

5

Technology & Digitalization

- Latest technologies – Woodgrove SFR for flotation
- Increased recoveries/reduced resource consumptions
- Digitalisation roadmap.

- ### Mining

 - ✓ First blast, Prestart mining commences July 15.
 - ✓ Mining contract outsourced – Aveng Moolmans Dec 16 .
 - ✓ North access ramp completion, Start of Bulk pre-stripping April 17.
 - ✓ Pre-stripping volume ramped up to 3.5 MT per month June 17.
 - ✓ Pre-stripping >25 MT of 65 MT completed as on July 17.
 - First ore production – Early 2018
 - Completion of pre-stripping, Bulk ore production – Mid CY 2018



- ### Plant

 - ✓ EPC order with EL Bateman Oct 16
 - ✓ Civil works progressing - Concrete bases for SAG and Ball mills completed Aug 17
 - ✓ All long lead equipment ordered and being manufactured. SAG & Ball mills dispatch Oct 17.
 - First ore feed, Start of production – Mid CY 2018.
 - 6-9 months for full production ramp up.



- ### Tailings Facility/Bulk Water

 - ✓ Tailing facility progressing on schedule (12 ha of HDPE liner installed against 110 ha total)
 - ✓ Water pipeline installation on schedule. 20 km of pipe delivered (6/38 km installed)

All major ordering with major commitments completed - Bulk mining, Plant & Infra and Tailings Dam . More than 80% of the capex committed. All contractors mobilised at site. More than 1,200 employees at construction site

- ### Other

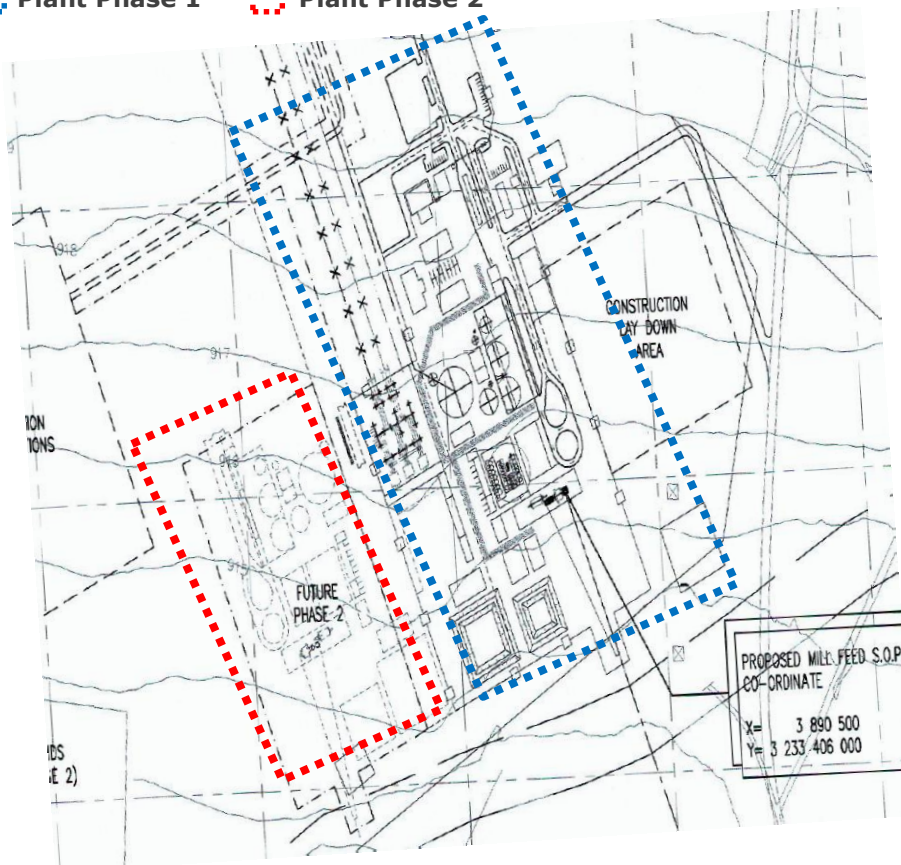
 - Biodiversity - ~85 000 plants relocated (2015)
 - Housing being constructed in anticipation of operational requirements
 - >100 houses being constructed

Gamsberg Phase 2 is Enabled by Phase 1

Phase 2 at Gamsberg increases the capacity from 4 MTPA to **8 MTPA** and reaches **450 KTPA metal in concentrate** capacity by developing a larger open pit mine, an additional 4 MTPA concentrator plant and associated infrastructure.

 **Plant Phase 1**

 **Plant Phase 2**



Phase-1 set up enables faster and capital efficient Phase-2 project execution

Regulatory approvals, biodiversity offset requirements and land readily available

Good understanding of the ore body

Mining infrastructure designed for expansion

Current **mine design/layout** incorporates the bigger open pit scenario. mine plan easily optimisable for 8 MTPA pit

Waste dump area, mining infrastructure already planned and designed for bigger pit

Outsourced model of mining operations gives flexibility to expand the mining production rapidly

All plant and infrastructure design considering the **modular approach** for easier expansion with minimum interface

Current layout considers additional **4 MTPA concentrator**

Water and power infrastructure available

Tailings dam – Needs only modular expansion

Technology and innovation used for setting up of the plant enables efficient scale up

Experienced project team already set up

Project execution approach – Outsourcing model with end to end responsibilities , business partners , already mobilised aware of the repeat orders

While the intention retrofit the Skorpion refinery to take Gamsberg concentrate, the volumes of Phase 2 may **warrant Gamsberg's own refinery**

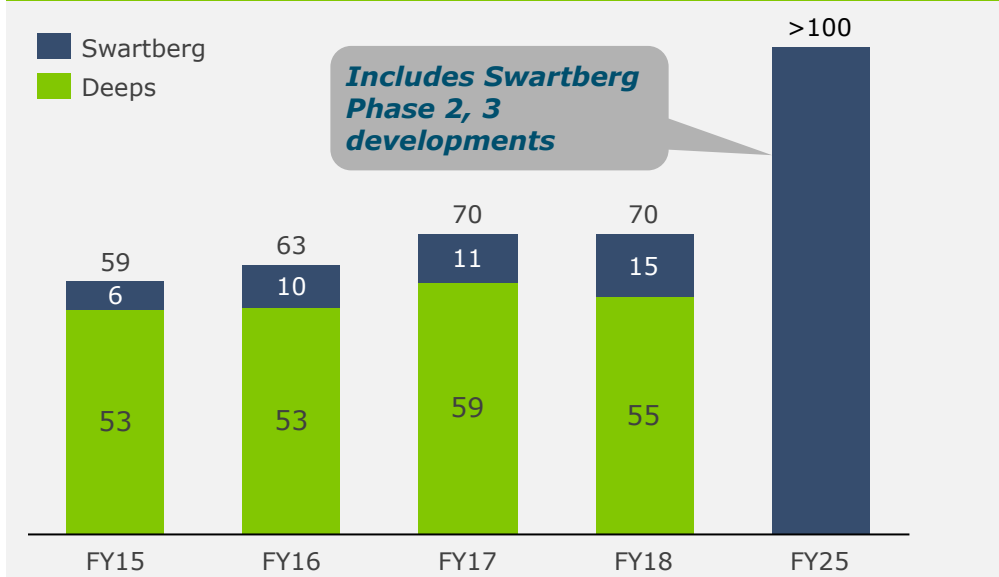
06

BMM – Enabling a world class Zn complex

Andre Trytsman - GM

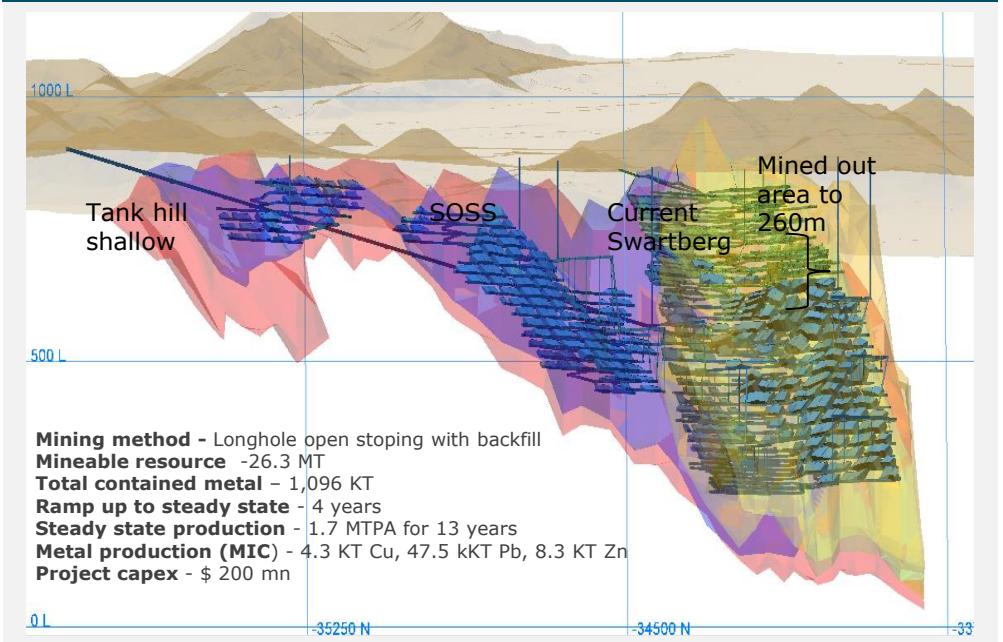
BMM Operational Performance is Trending Positively...

Production per year, Metal (Zn & Pb) in concentrate



- BMM has performed exceptionally over last 4 years, despite:
 - Decreasing working areas (from 10 to 5)
 - Increasing workplace congestion
- Specific initiatives over last 2 years include:
 - Backfill placement up by >80%
 - Long hole mining up by >190%
 - Mining productivity up by 10%
 - Milling throughput up by 11%

..and can be Further Enhanced with Swartberg Phase 2



Black Mountain Complex – Operational readiness

- Strong Future complex moving toward 500kt ZnEq - Gamsberg 450kt, Swartberg 75Kt
- Synergy benefit on combined Black Mountain Complex (BMC) - Shared overhead structure, Commercial and Shared Infrastructure
- Combined operating model - outsourcing of Gamsberg vs owner operation on current operation. Opportunity to ensure optimal operation.
- Efficiency improvements via higher level of automation and digitalisation.
- Establishing a world class Zinc complex increases our ability to attract and retain the best talent.

07

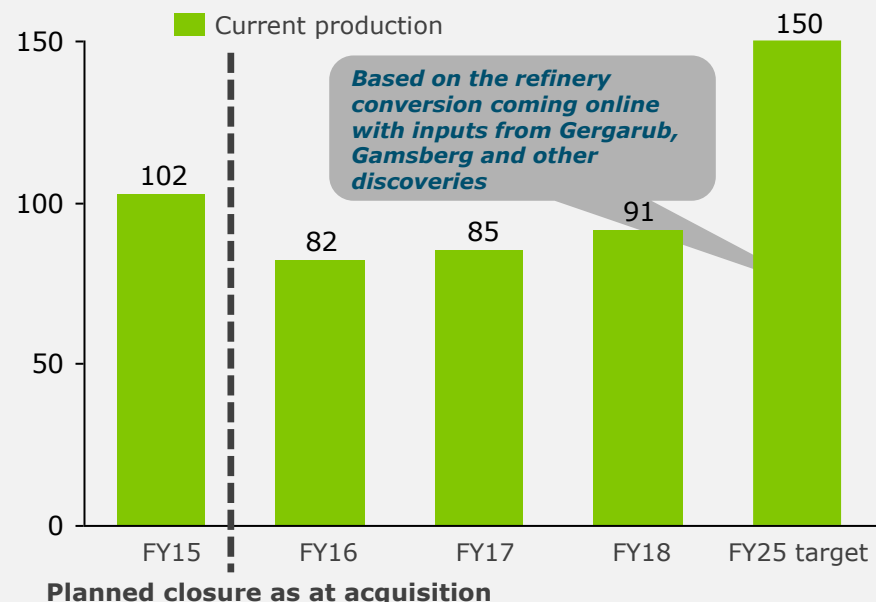
Skorpion - Surpassing LOM expectations

Irvinne Simataa - GM

Skorpion Zinc's Increased LOM Certainty & U/G Potential

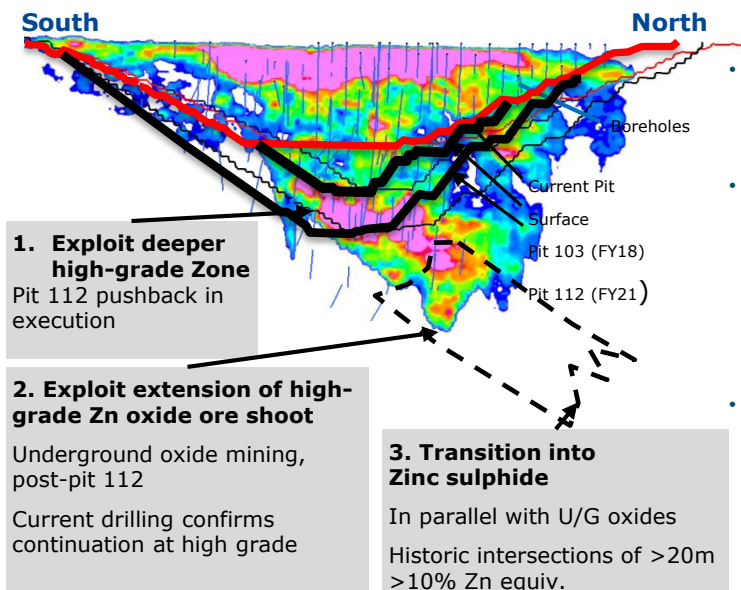
The delivery of Pit 112 has created more certainty around SZ....

Production per year, Zn metal



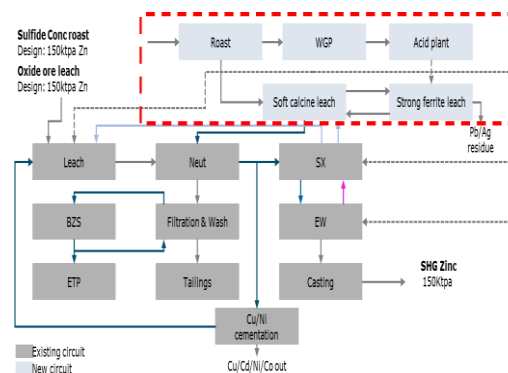
- Skorpion has continued to deliver due to:
 - Extension of life from 0.5 to 4 years via the Pit 112 pushback
 - Successfully outsourced mining to Basil Read Namibia in April 2017 and transitioned from owner mining to contractor mining to enable successful execution on the life extension project
 - Mining productivity up by 33% and total rock mined has increased by 83% YoY
 - Continued investment in plant maintenance (e.g. acid plant, crushing etc)
 - A targeted operational improvement programme to further enhance throughput and recovery within the refinery and reach nameplate capacity of 150 000 tpa

...With further potential at Pit 112 and the planned refinery conversion



- Currently mined Skorpion orebody highly variable in grade and geometry
- Orebody bulges out at depth, hosting a significant accumulation of metal (hence Pit 112), resource still open at depth (hence UG study).
- Average Zn grades as high as at beginning of LOM.

Refinery conversion



Current Status

- Current refinery need to be converted to process sulphide concentrates.
- Techno commercial feasibility/Basic Engineering completed.
- ~160 million USD Capex.
- Project ordering & construction work timelines aligned to Skorpion open pit 112 extension (oxide resource)

08

Digitalization & Technology

- Driving efficient growth

Dave Payne – Head of D&T

Enabling Growth Through Digitalisation

The Digitalisation of Gamsberg is the step change in our journey and will be the springboard from which we can transform the rest of our business

BMC Initiatives

Change of Mining Method from predominantly C&F to Long Hole



Latest mining equipment



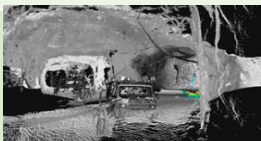
Automated Rod Charger



In-hole survey (Gyro)



Maptek Isite 3D laser scanner



BMC Magnetite Project



SZ Initiatives

Screening of high Ca Stockpile ROM to reduce calcium content



Optimising the ROM ore blending strategy



Change from Shovel to Backhoe for ore mining



Skorpion Marginal Grade Stockpile

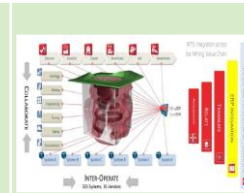


Skorpion Cu Stockpile



Gamsberg Digitalisation Roadmap

Collaborative Ore Movement



- Increase flexibility & execution of mine plan
- Enhanced contractor management
- Increased control on mill feed

Asset Intelligence



- Increase reliability/availability /utilisation
- Reduce maintenance costs
- Increase production

Operational Intelligence



- Stabilise & optimise operations
- Increase throughput & recovery
- Near real time decision making

Personnel Monitoring



- Increase safety & reduce incidents
- Increase efficiency
- Increase quality & production

09

Financial outlook – The path forward

Pushpender Singla – CFO

Reserve & Resource >300Mt Ore

High value Resource Base 306Mt (22Mt Metal) to enable long term sustained Production and Growth

- R&R on acquisition 286Mt. Addition of 44Mt (>15%) including depletion
- Converted 63Mt Resource to Reserve (>20%)

Skorpion & Black Mountain

>200Kt Production

- **FY2018: Production of c.160 KT @ < CoP of \$ 1,500/t**
 - Skorpion c.90 Kt & Black Mountain c.70 Kt
 - Focus on further reducing COP through operational and cost initiatives
- **FY19 & FY20: Plan to increase production to c.200 Ktpa by ramping up Pit 112**
 - Skorpion c.130 Kt & Black Mountain c.70 Kt

Gamsberg

250Kt Production@ ~\$1000/t

- **Project on track and progressing as per plan**
 - First production by mid CY2018 and on budget for capex target of \$400 mn
 - FY19 production expected to be 100 Kt+, with ramp-up to 250 Kt by FY2020
 - Targeted COP of \$1,000-1,150/t, placing ZI in 1st quartile of global cost curve

Indicative Consolidated Outlook

	FY18	FY19	FY20	FY21	FY22
Prod (Kt)	150-160	240-260	410-430	430-450	440-460
COP (U\$/t)	<1500	<1200	<1100	<1000	<900

<https://youtu.be/KKpDTXCszGU>



Vedanta - Zinc Day

Hindustan Zinc Limited

30 August, 2017



OUR ZINC PROTECTS
STRUCTURES FOR
FUTURE GENERATIONS



AND GENERATIONS
FOR OUR FUTURE



01

Corporate Overview

Amitabh Gupta, CFO

02

Social Responsibility – Giving Back

Neelima Khetan, Head CSR

03

Delivering 1.2 mtpa

Naveen Singhal, Director Projects

04

Resource-Driven Growth

Scott Caithness, Head Exploration

05

Silver – Fastest Growing Portfolio

Sunil Duggal, CEO

06

Way Forward

Sunil Duggal, CEO

Q&A



01

Corporate Overview

Amitabh Gupta, CFO



2nd largest zinc-lead miner globally with
12 million+ MT of ore production capacity

Rampura Agucha: **2nd** largest zinc mine globally

4th largest zinc-lead smelter globally: **1 million+ MT**
of production capacity

18th largest silver producer of the world*

One of the lowest cost zinc producers in the world

3 mines with **five-star** ratings by Indian Bureau of
Mines

Vision: To be the world's largest and most admired zinc-lead & silver company

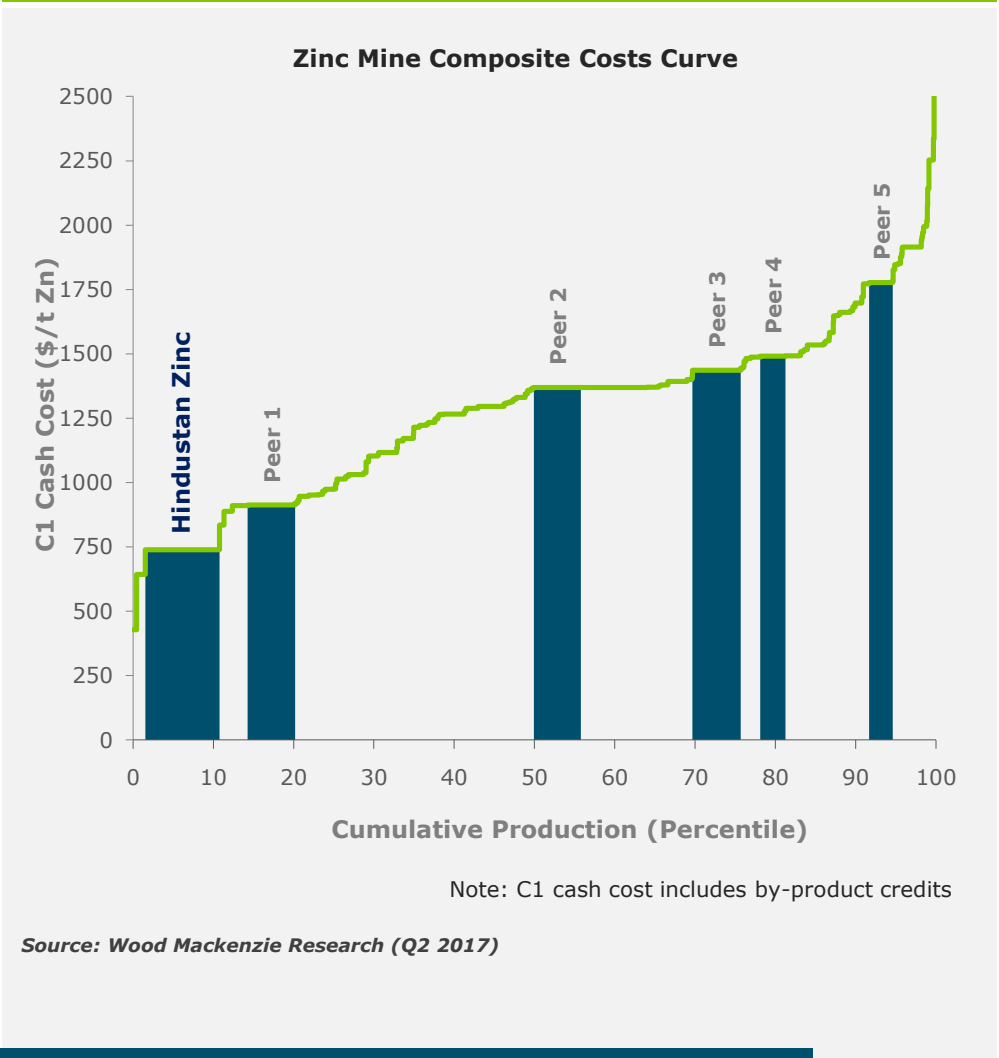
Rankings source: (1) Wood Mackenzie, Long-Term Outlook Q2 2017
(2) For HZL related data, actuals for FY 16-17
* Source : Thomson Reuters



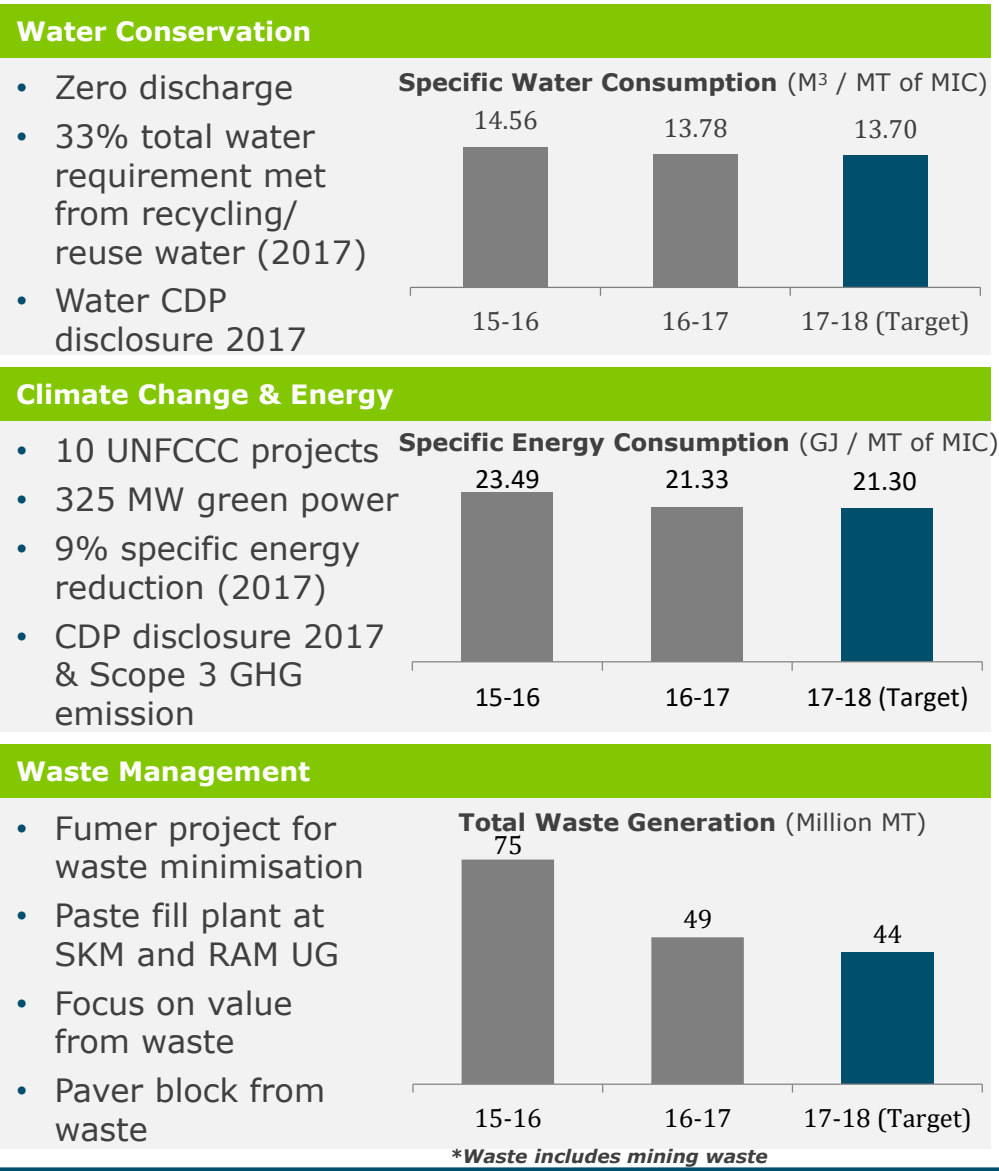
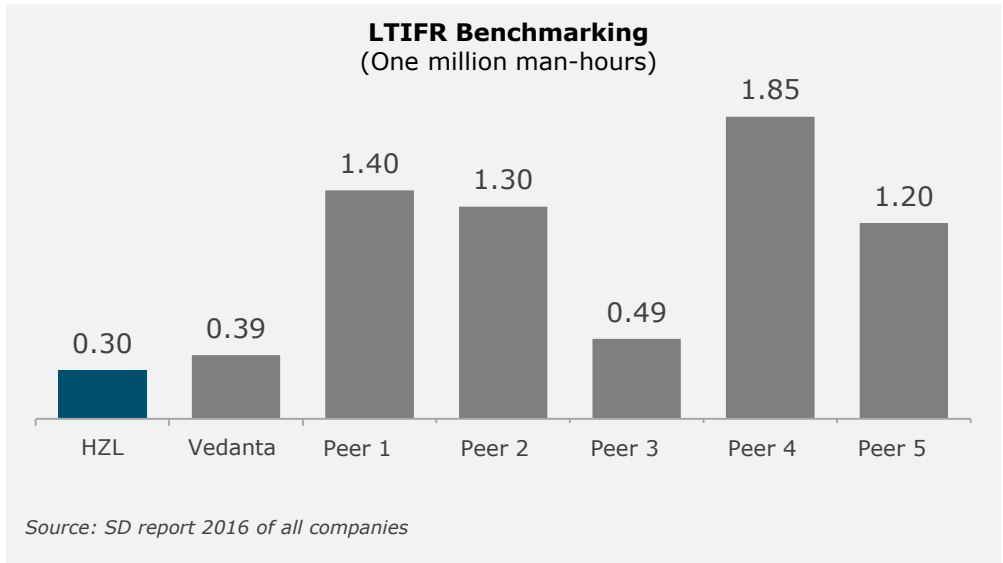
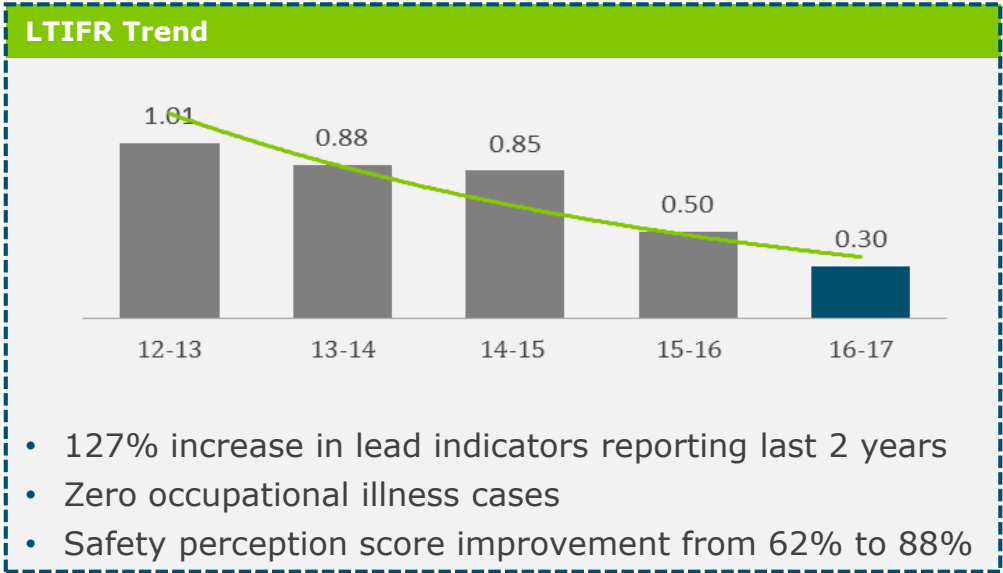
Core Strengths

- 1** Captive mines with R&R base of 404.4 million MT ensuring mine life of 25+ years
- 2** Integrated metal production supported by captive power plants
- 3** Low cost of operations driven by quality assets and recovery from waste
- 4** Market leadership in India with strong presence in emerging economies of Asia

Among the Lowest Cost Producers of Zinc



Leading the Way with Fully Integrated Operations





Rampura Agucha Mine

Reserves : 49.7 mn MT
Resources : 50.3 mn MT
Reserve Grade : Zn 13.9%, Pb 1.9%
Current Ore Capacity : 6.15 mtpa

Kayad Mine

Reserves : 5.4 mn MT
Resources : 1.7 mn MT
Reserve Grade : Zn 7.8%, Pb 1.1%
Current Ore Capacity : 1.00 mtpa

Sindesar Khurd Mine

Reserves : 35.6 mn MT
Resources : 87.2 mn MT
Reserve Grade : Zn 4.2%, Pb 2.9%
Current Ore Capacity : 4.50 mtpa

Rajpura Dariba Mine

Reserves : 9.0 mn MT
Resources : 50.3 mn MT
Reserve Grade : Zn 6.3%, Pb 1.5%
Current Ore Capacity : 0.90 mtpa

Zawar Mining Complex

Reserves : 9.5 mn MT
Resources : 85.7 mn MT
Reserve Grade : Zn 3.3%, Pb 1.8%
Current Ore Capacity : 4.00 mtpa
CPP : 80 MW

Chanderiya Smelting Complex

Pyrometallurgical Lead Zinc Smelter:
1,05,000 MTpa zinc and 35,000 MTpa lead
Hydrometallurgical Zinc Smelter:
4,20,000 MTpa zinc
Ausmelt™ Lead Smelter: 50,000 MTpa lead
Captive Power Plant: 234 MW

Dariba Smelting Complex

Hydrometallurgical Zinc Smelter:
2,10,000 MTpa zinc
Lead Smelter: 1,00,000 MTpa lead
Captive Power Plant: 160 MW

Zinc Smelter Debari

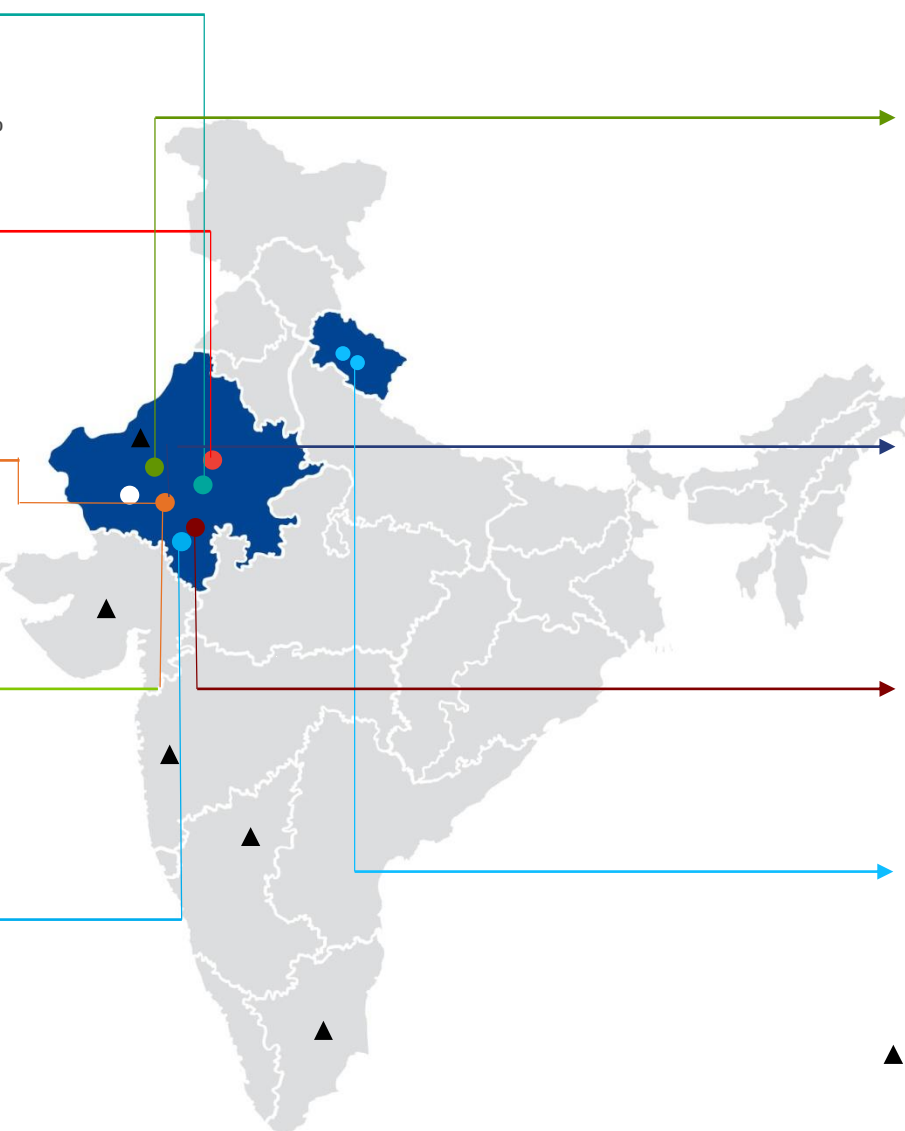
Hydrometallurgical Zinc Smelter:
88,000 MTpa zinc

Pantnagar & Haridwar

Processing and refining of zinc,
lead and silver

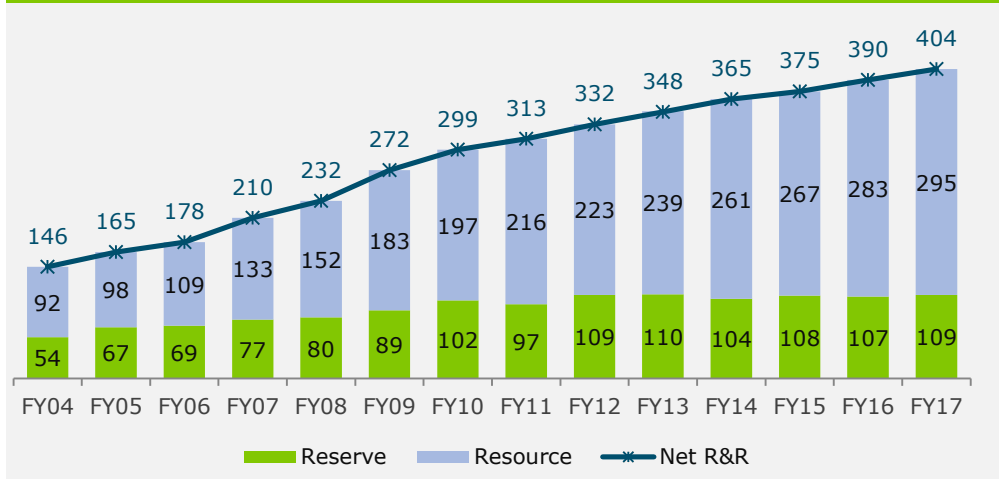
Renewable Power

▲ **Wind:** 274 MW
Solar: 16 MW
Waste heat recovery: 35 MW

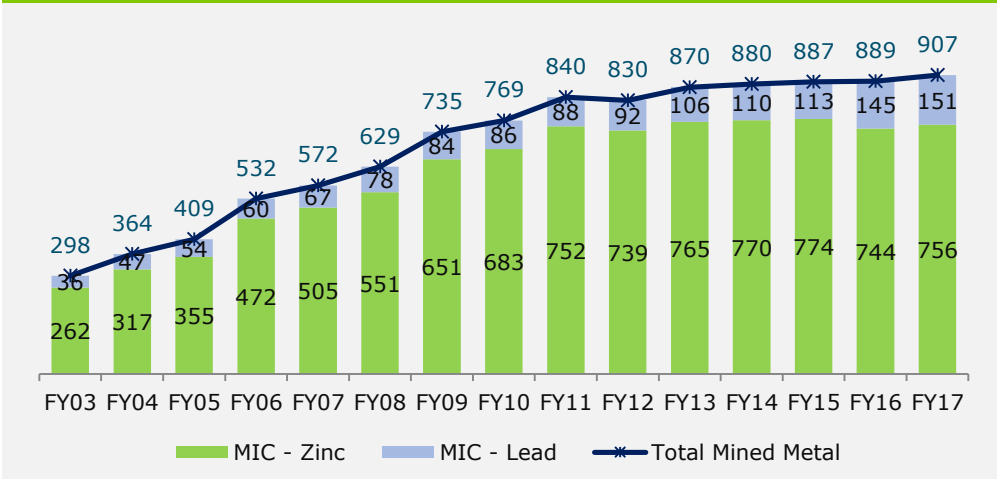




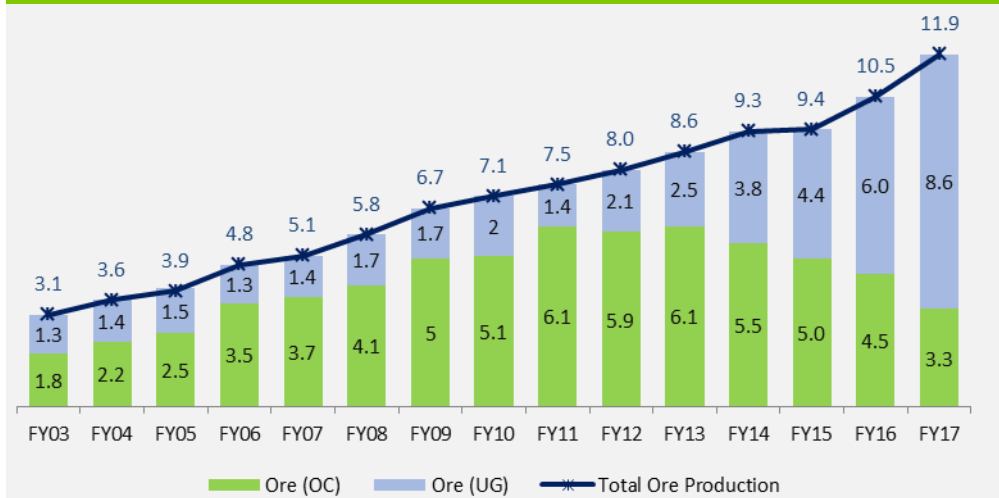
Reserves & Resources (mn MT) – 2.5X Growth



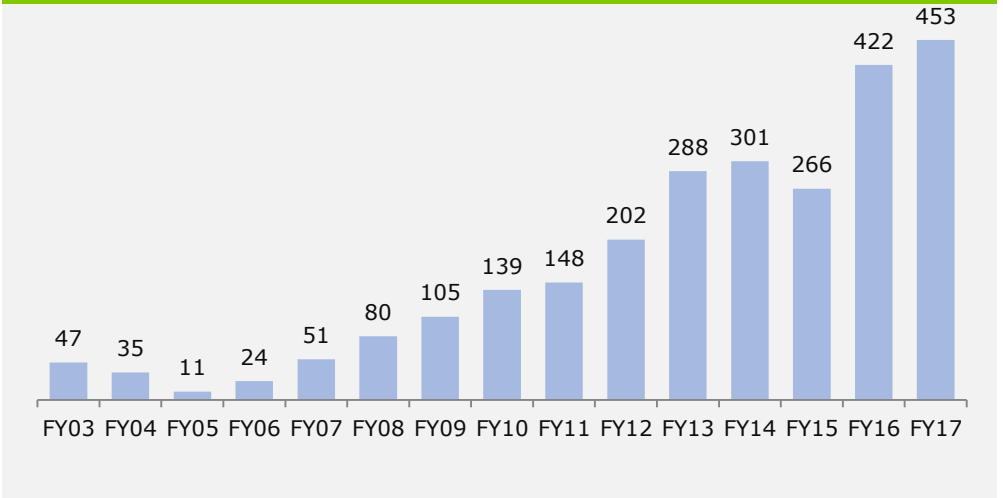
Mined Metal (kt) – 3X Growth



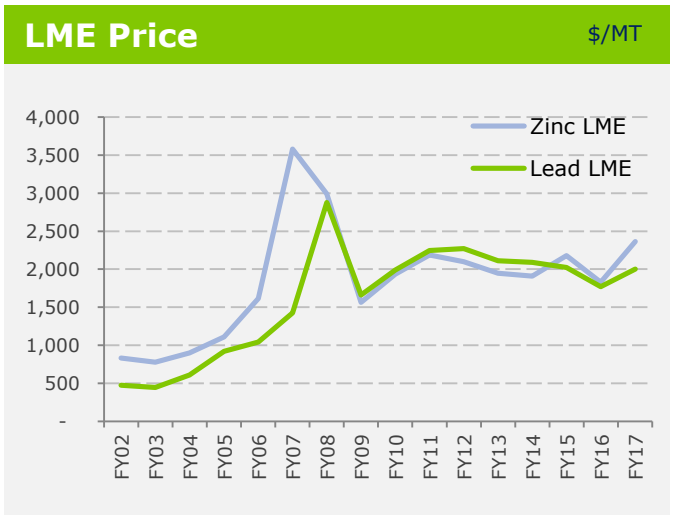
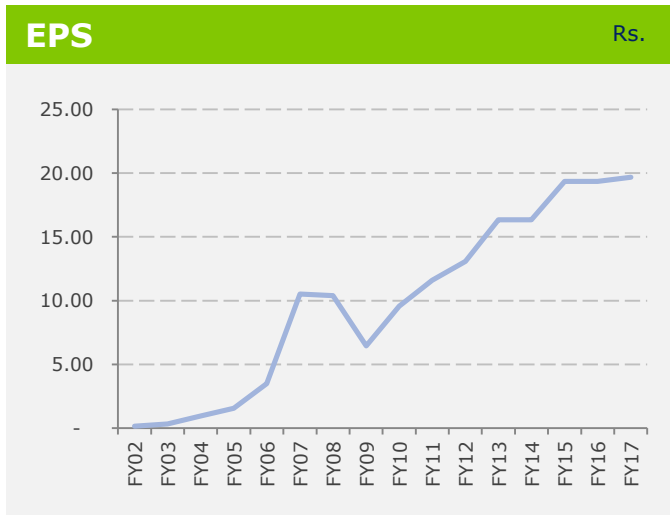
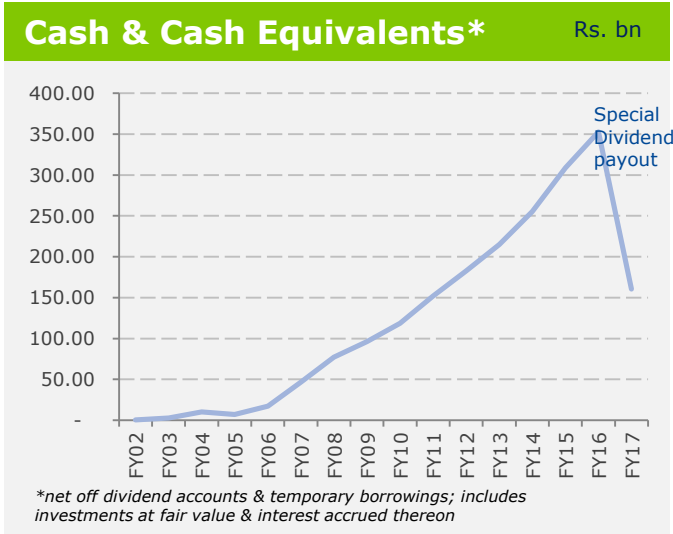
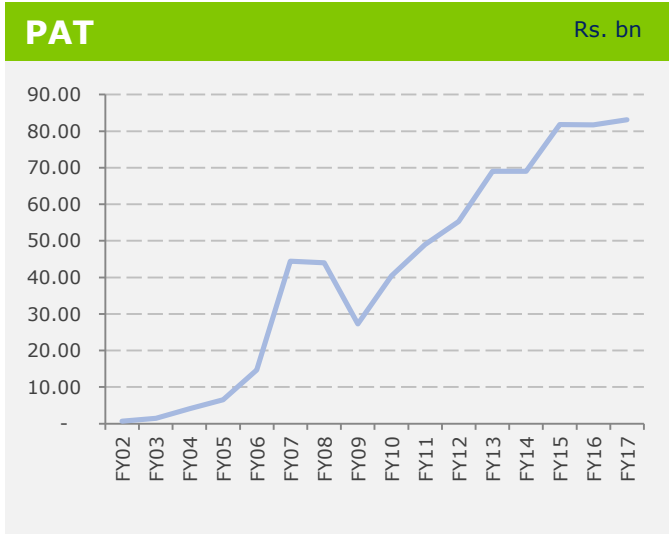
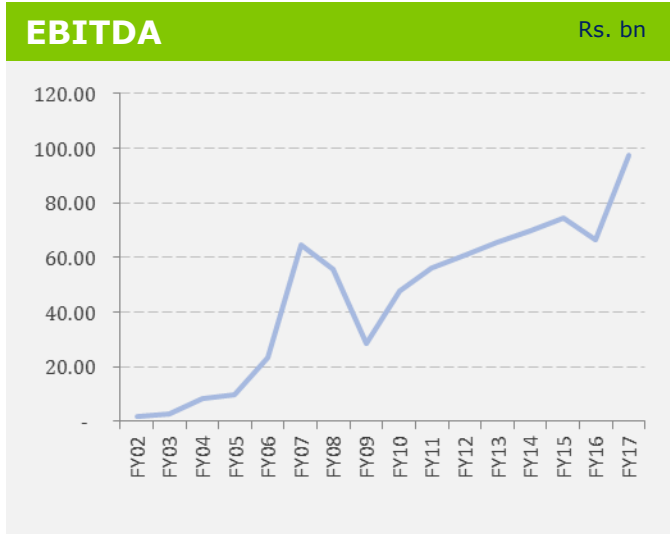
Ore Production (mn MT) – 3.5X Growth



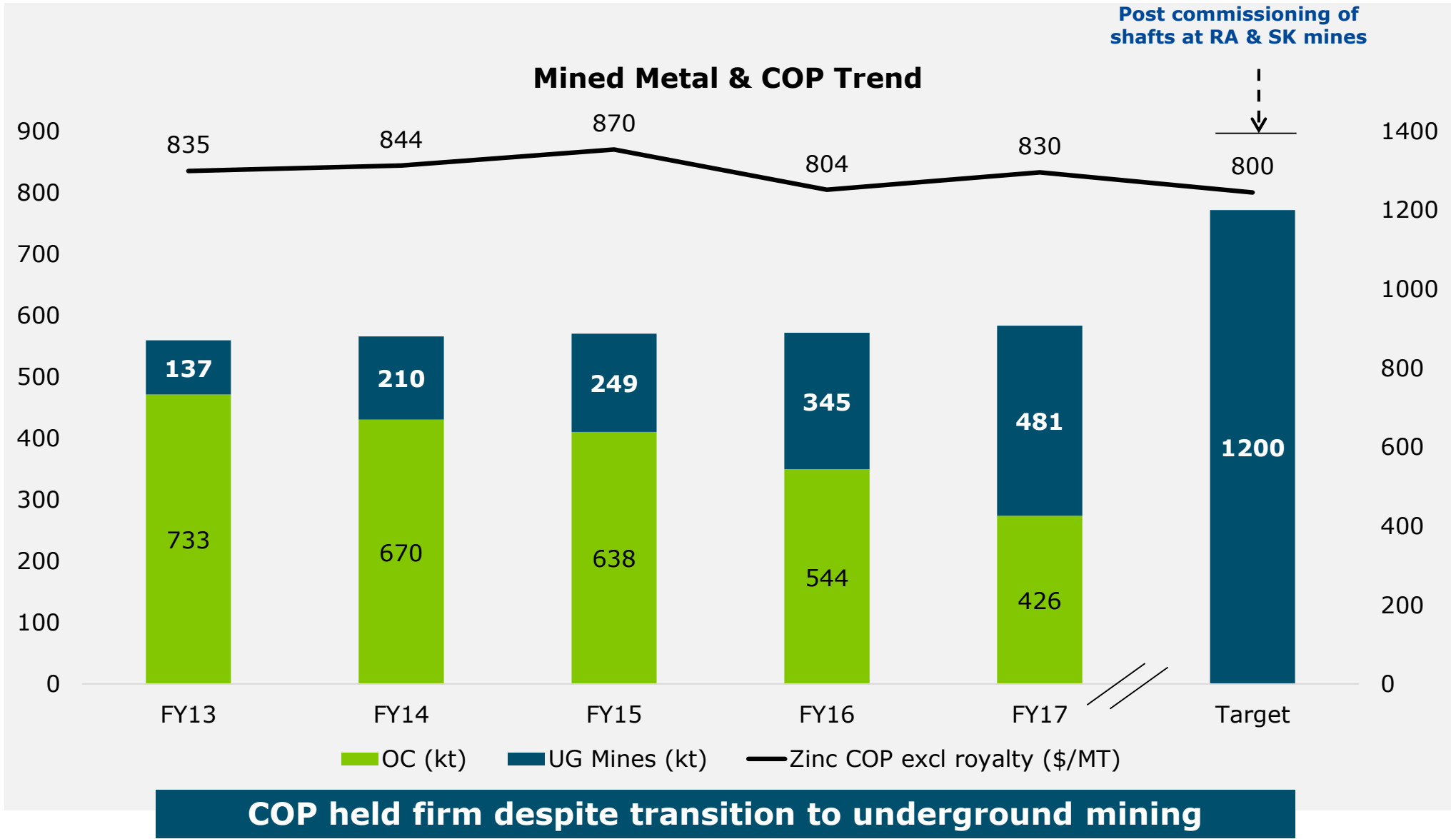
Silver Metal – Integrated Saleable (MT) – 9X Growth



Multi-fold growth since disinvestment



EBITDA CAGR 28% despite Zinc LME growth of only 7%





02

Corporate Social Responsibility

Neelima Khetan
Head CSR



Vision

To enhance the quality of life and economic well being of communities around our operations.

Goals

- To positively & holistically impact the quality of life of the communities living around our areas of operation.
- To work in partnership through a multi-stakeholder approach for innovatively, effectively and efficiently addressing development challenges.
- To emerge as a thought leader, creating benchmarks of good practices in CSR across Rajasthan & country.

Reach

Over half a million people in Rajasthan; positively impacting lives of more than 100,000 households

HZL is proud to have had an unbroken social license to operate for the last 5 decades

Core Principles

Neighbourhood Communities

Our first and foremost commitment

The Partnership Multiplier

Work, as much as possible, through strong and credible local partners

Impact Centric

In almost everything we do, we focus on impact rather than on spends

Strong Governance

Multi-tiered and objective governance structures and routines

Thematic Areas



Education



Sustainable Livelihoods



Women's Empowerment



Health, Water & Sanitation



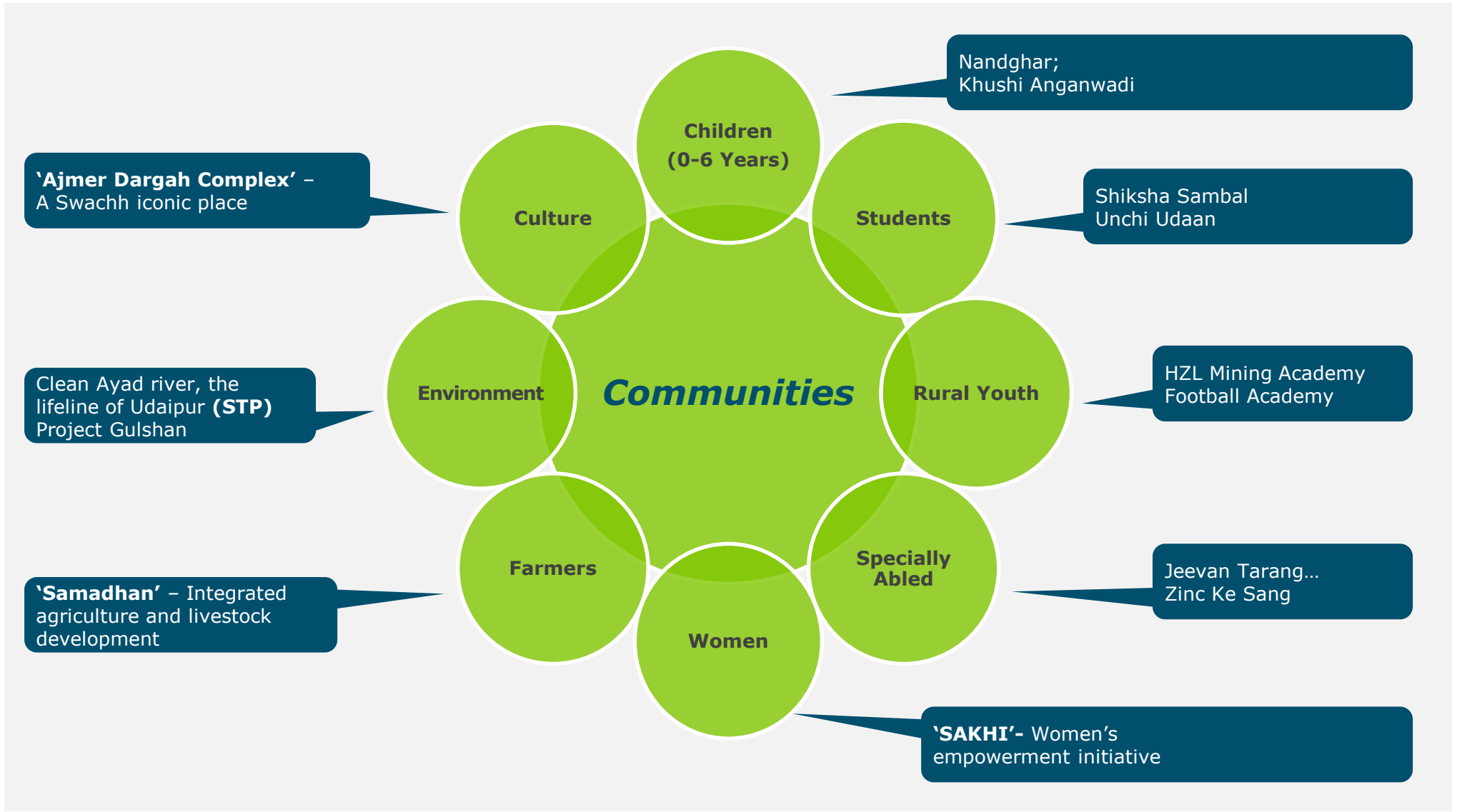
Sports & Culture



Environment



Community Assets Creation

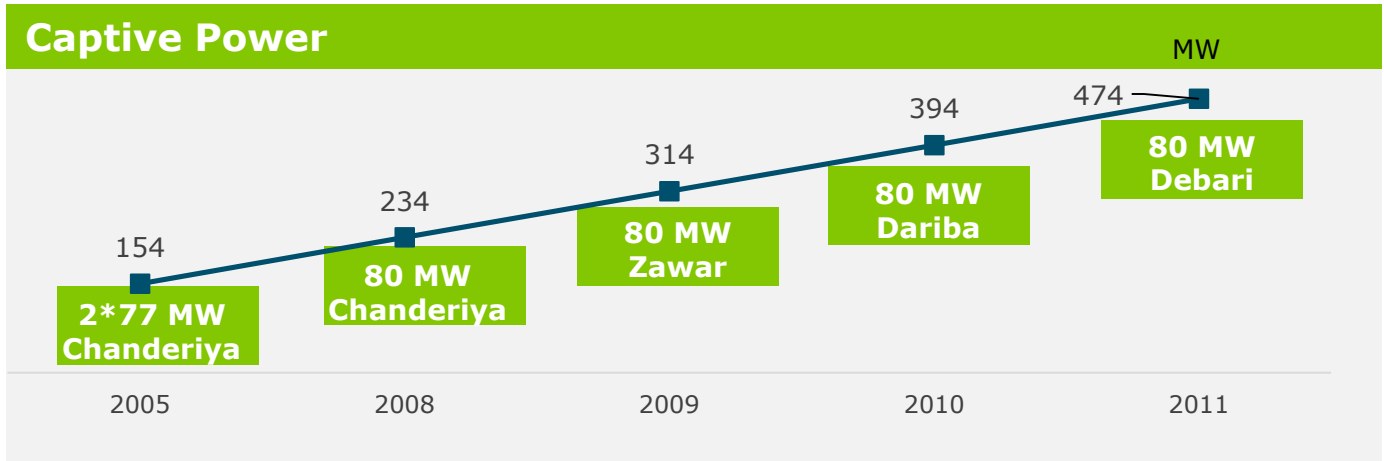




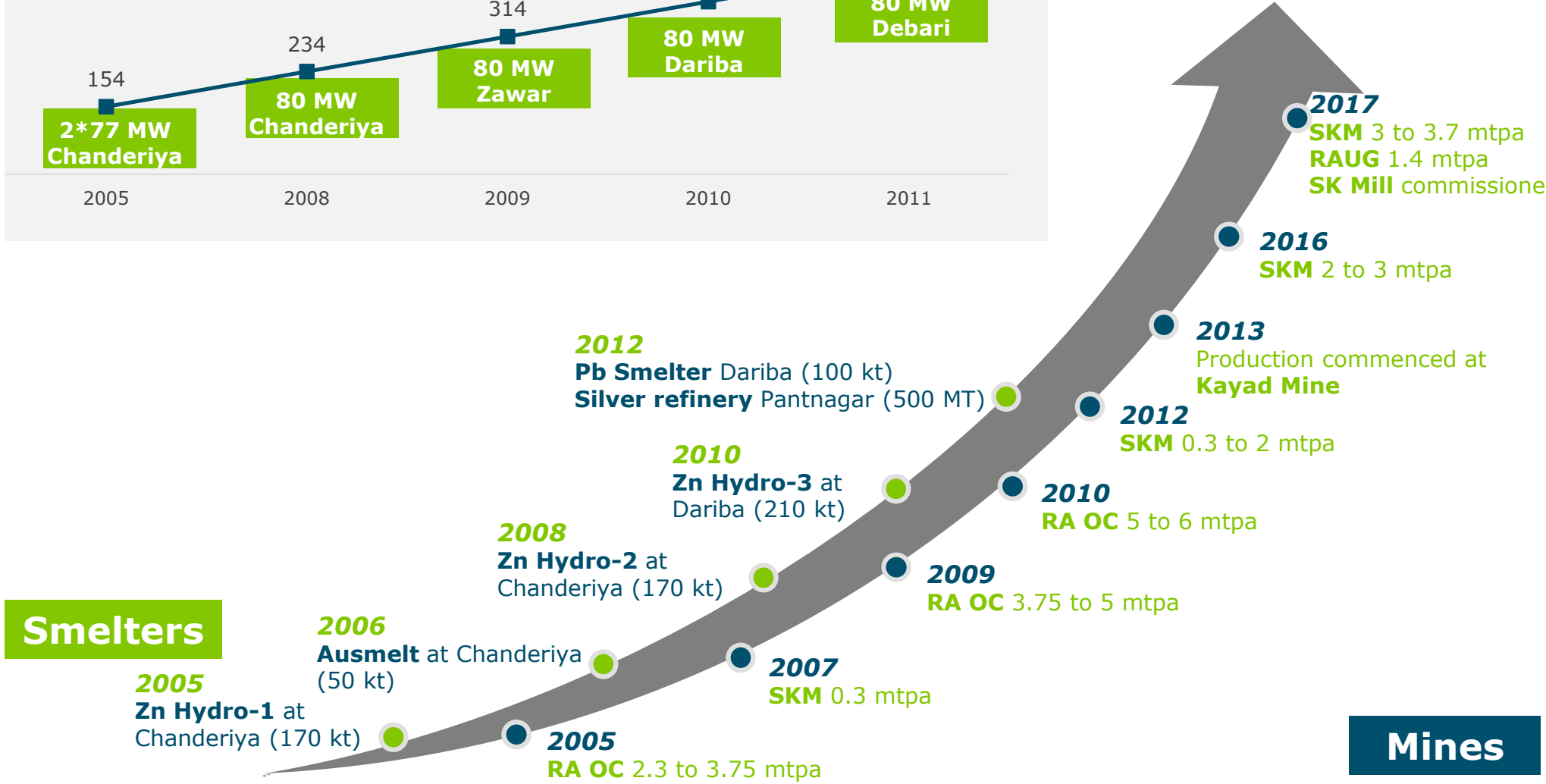
03

Delivering 1.2 mtpa

Naveen Singhal
Director Projects



Smelters



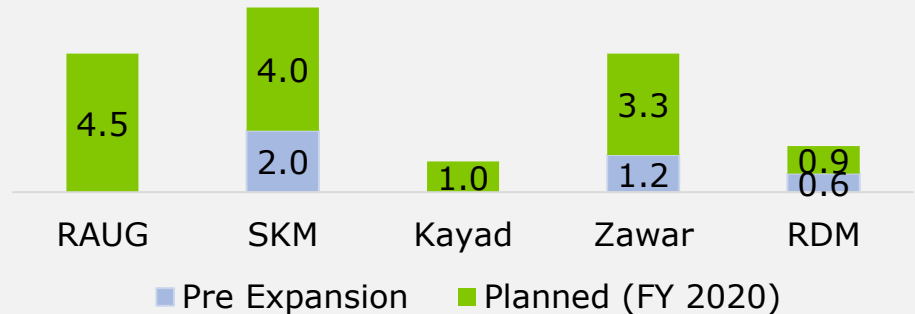
Mines



- Mine expansions in sync with increasing R&R
- Five major projects to increase MIC capacity to 1.2 mtpa, including RAM OC replacement
- Transition from Open Cast to UG mining
- Smelters' debottlenecking to remain integrated
- Six year capex of \$1.6 bn; ~\$1 bn spent

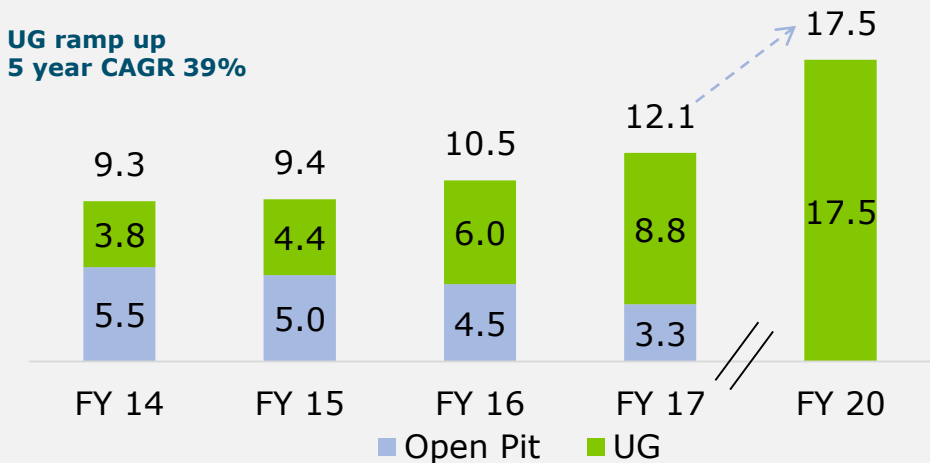
Underground Mines Capacity Expansion

3.8 mtpa → 17.5 mtpa



OC to UG Transition

UG ramp up
5 year CAGR 39%

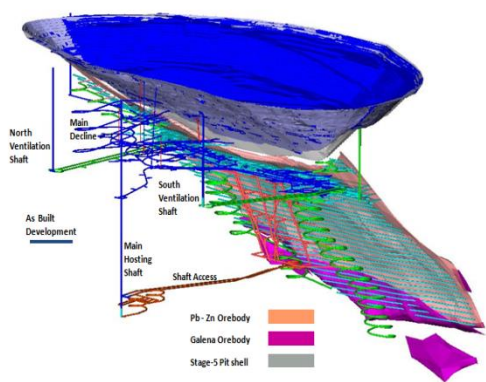


Underground Mines – concurrent expansion



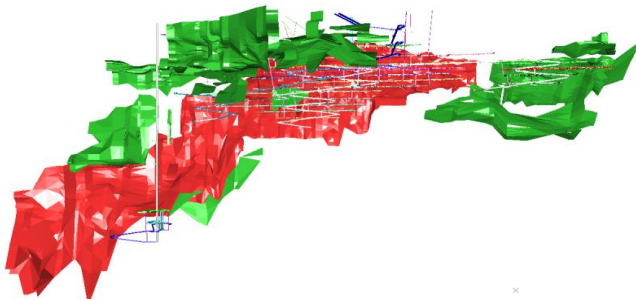
Rampura Agucha – a mine in transition

Reserve & Resource : 100.0 mn MT



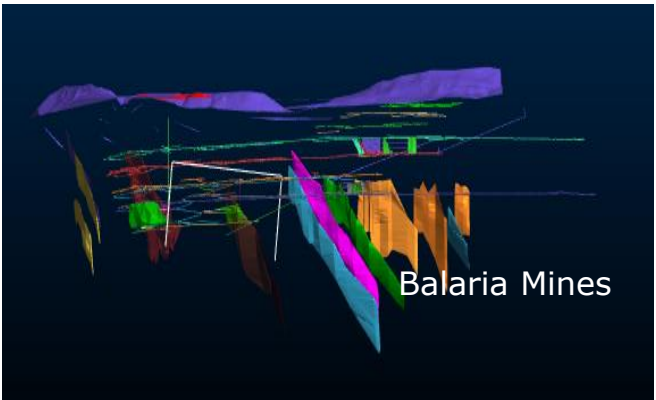
Sindesar Khurd Mine – satellite lenses

Reserve & Resource : 122.8 mn MT



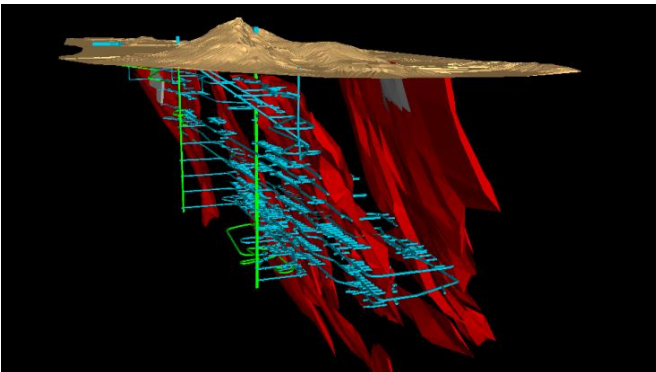
Zawar – integrating multiple mines

Reserve & Resource : 95.2 mn MT



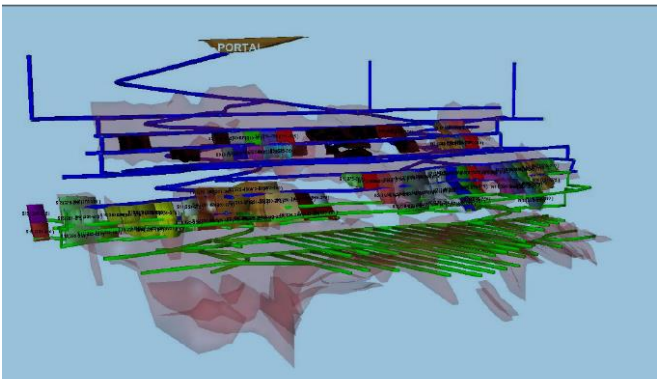
Rajpura Dariba Mine – upgrading a legacy

Reserve & Resource : 59.3 mn MT



Kayad Mine – near mine exploration

Reserve & Resource : 7.1 mn MT



Diverse ore bodies require different mining solutions



Robust Mine Plan

- LOMP, Mining schemes by AMC, Australia and reviewed by Golder, Canada
- Rock Mechanics & Stability Study by AMC - Beck, Australia
- **Central Design Center** - act as a think tank for adapting best mining design / practices



World-Class Infrastructure

- **Safety** - Rescue chambers, Gas Leak Detectors, Emergency escape systems
- **Communication** - Leaky feeder systems, Pitram mine Monitoring
- **Ventilation** - Development of Independent ventilation circuit by using Raise bore
- **Shafts** - 7.5 mtrs dia , 1 km deep shaft for man, material & ore hoisting
- **Workshop** - Fully equipped at surface & Underground for equipment maintenance by OEM
- **Pastefill** - Most Modern and efficient back filling



Best-in-Class Execution Partners

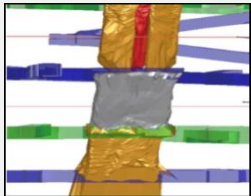
- **Shaft Sinking** - METS - Shaft Sinkers, South Africa; NFC- ENFI - CC5, China
- **Mine Development** - Barmenco, Australia; AAC - Reliant, Peru; Indian contractors, Expats
- **Mining Fleet** - Atlas Copco - Sandvik - Normet, Sweden; Catter Pillar, USA
- **Raise Boring** - Bergteamet, Sweden; Atlantis, South Africa
- **Paste Fill Plant** - Golder Associates, Canada



Talent Pool

- **Experienced expats** on board for handholding, mentoring & training HZL team
- **Mining academy** - Skill development and enhancement of local youth in UG operation
- **Skill development** and competency building of existing manpower pool

Technology Initiatives



Cavity Monitoring System: Stope scanning identifies and minimises deviation



Refuge Chamber: First of its kind in the country, provides a safe refuge during UG emergency

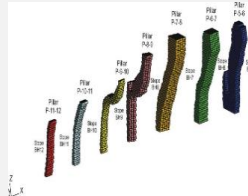


Mobile Carrier Rig: Eases mobility of exploration drills



1500 KW Ventilation Fan: First of its kind in the country, provides best UG working environment.

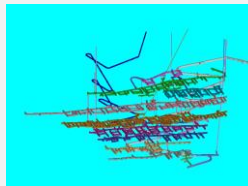
Mine Planning



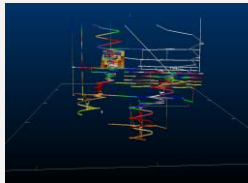
Measurement 3D numerical modelling for stope and pillar design



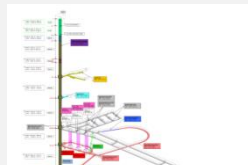
In-situ stress measurement



3D design using datamine



Schedule simulation using 5D planner



Shaft sinking modelling

Mine Operation



Best-in-class mining equipment



Virtual reality and simulation-based training



Mine digitisation and tele remote



3D scanning with laser scanner



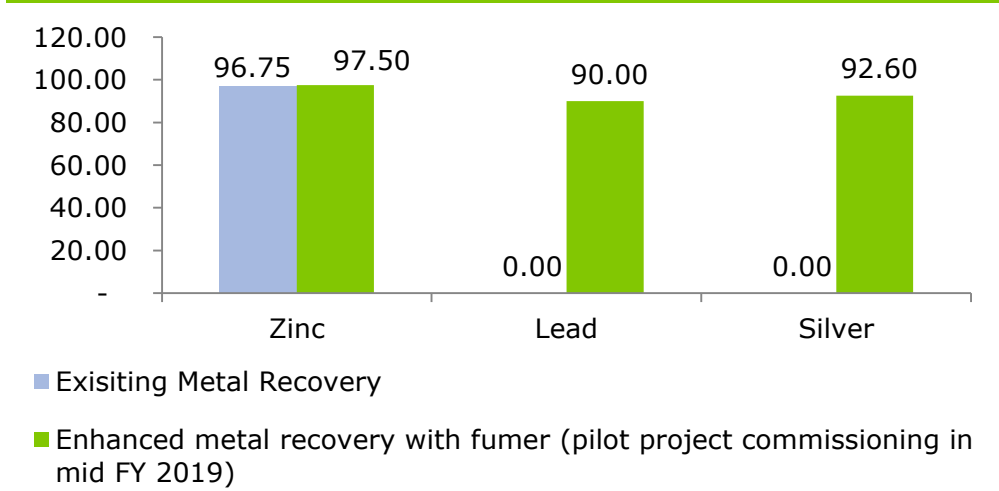
Electronic shot firing

Accelerated focus on safety & productivity leveraging technology

Process & Technology

Metal	Process	Technology
Zinc	Hydro Metallurgy	Outokumpu roast-leach-electro winning technology
	Pyro Metallurgy	Imperial smelting technology
Lead	Pyro Metallurgy	Imperial smelting technology
		Ausmelt Top Submerged Lance technology
		SKS, bottom blowing technology

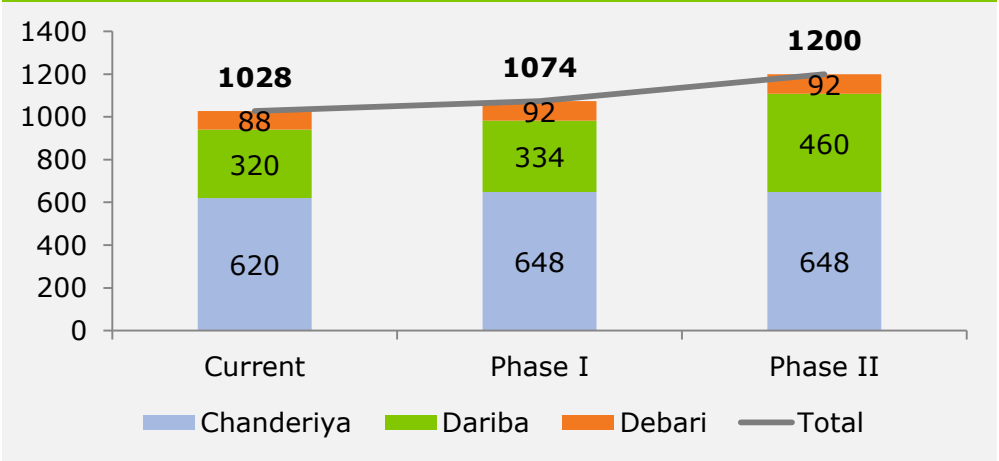
Metal Recovery in Hydro Metal recovery %



Smelter Debottlenecking

- Phase 1 : December 2017**
- Cell house operation @ 200 KA from 180 KA
 - Upgradation of Roaster leaching circuits
 - Mechanisation of cell house at Debari zinc plant
- Phase 2 : December 2018**
- Expansion of Dariba zinc cell house
 - Leaching & purification circuit debottlenecking @ Dariba
 - Oxygen enrichment in roasters to increase throughput

Smelter Capacities After Debottlenecking kt





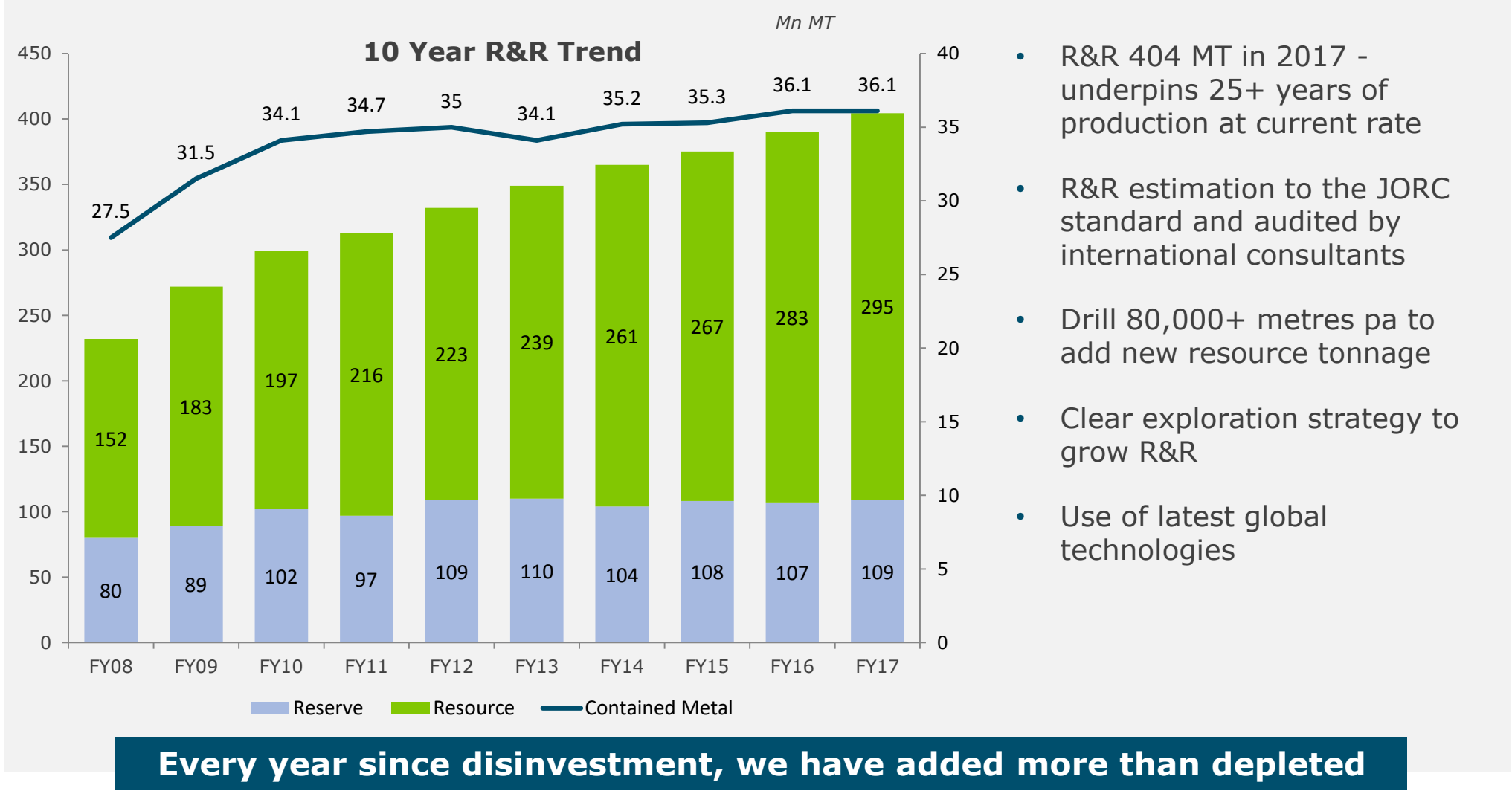
04

Exploration

Scott Caithness
Head Exploration



Sustained Commitment to Exploration → Strong Track Record of R&R Addition



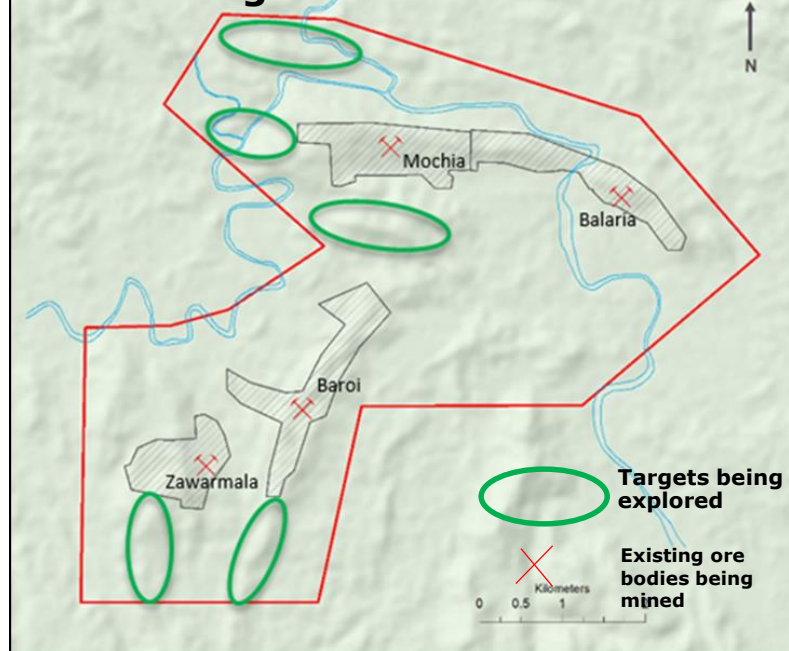
Resource-Driven Growth

- All deposits remain 'open' either at depth or along strike
- New exploration targets could exist within held MLs
 - Zawar in-mine bonanza silver – 2-8 metres zones grading +5oz/t Ag
 - New prospects in all MLs
- Zawar has 5 new prospects
 - Exploration potential for 50-80 mn MT of 4-6% Zn+Pb
 - +5km of prospective geology, strong Zn+Pb in soils and ancient mine workings
 - Next steps - Surface geophysics, forest clearance and drilling
- More than 20 valid ML, PL and RP applications provide potential for greenfield discoveries

Zawar Bonanza Silver

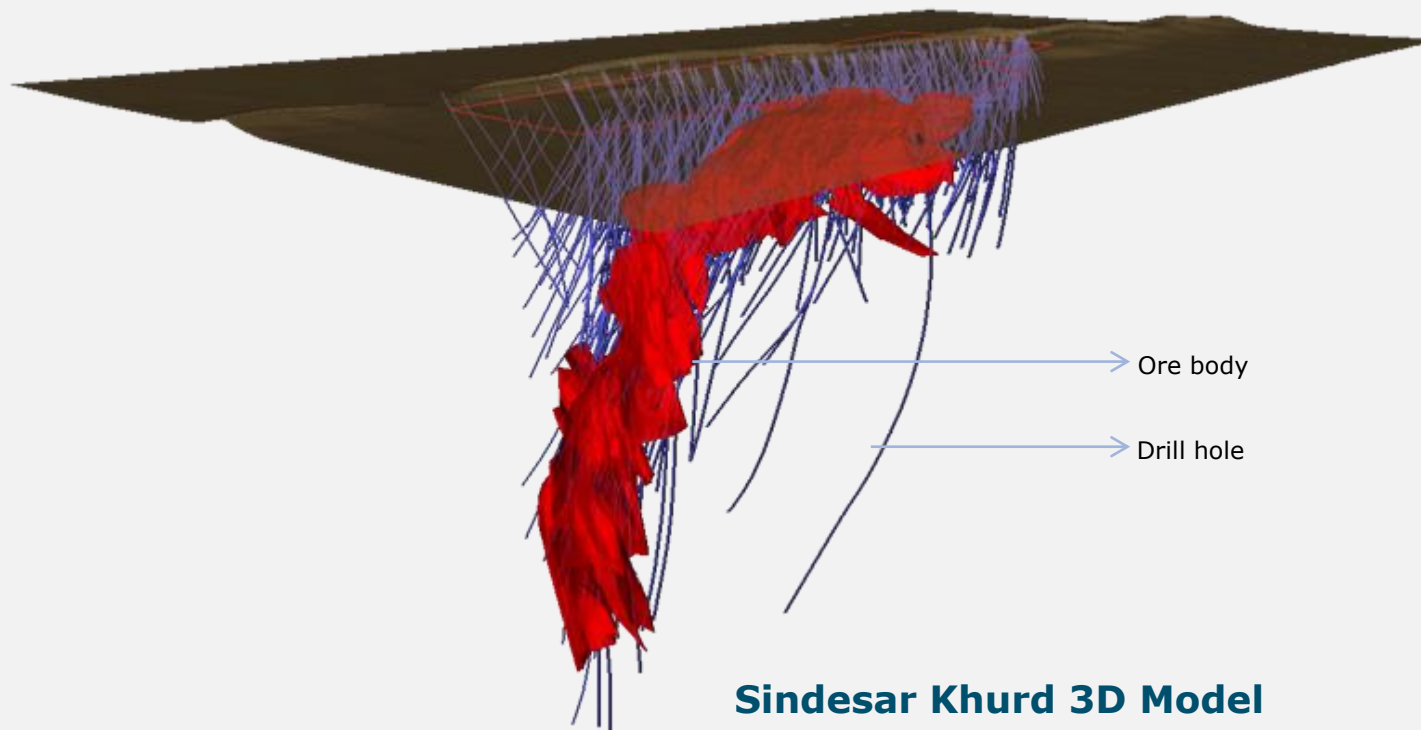


Zawar Targets



Using Global Leading Exploration Technologies

- Modern drill rigs and motorised directional drilling
- Drilling and geochemical data management software
- 3D visualisation software packages
- Geophysical techniques



**High speed drill rig at
Kayad Mine**



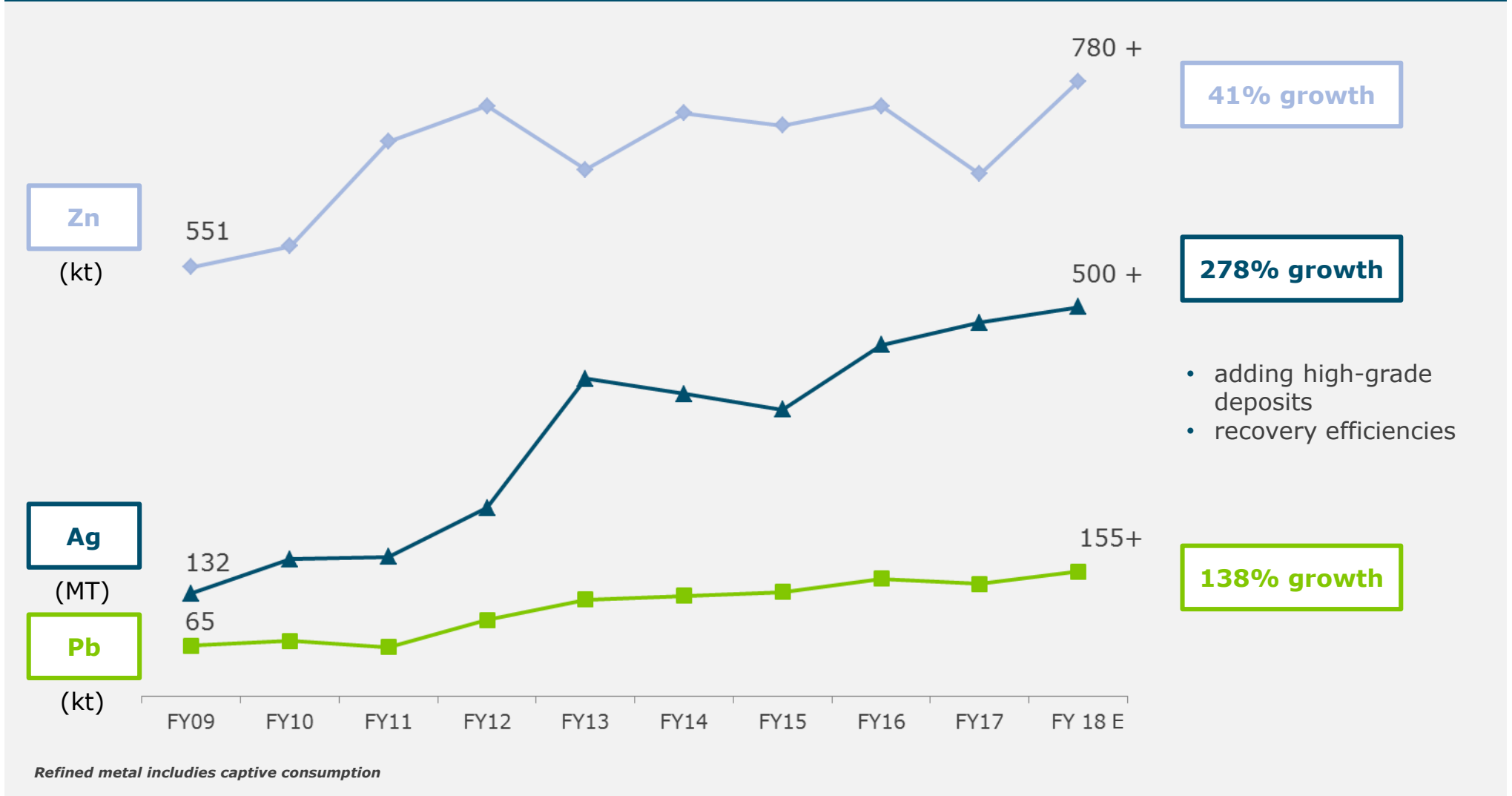
05

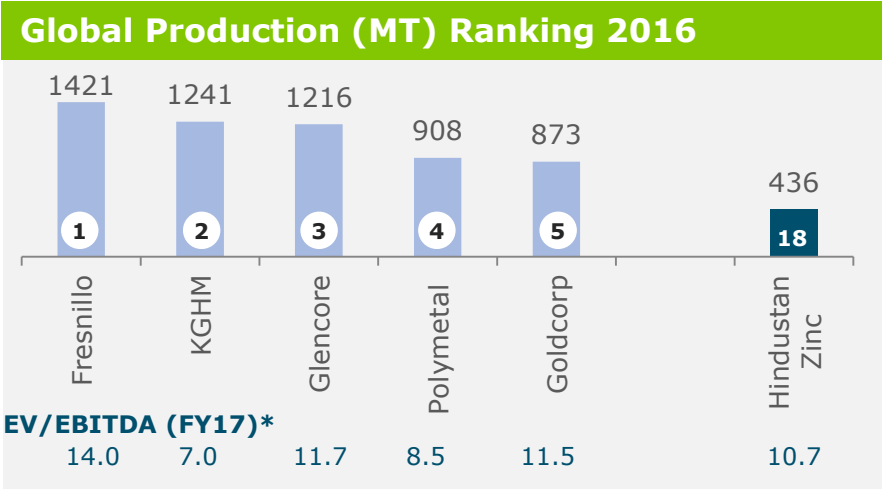
Silver

Sunil Duggal
CEO

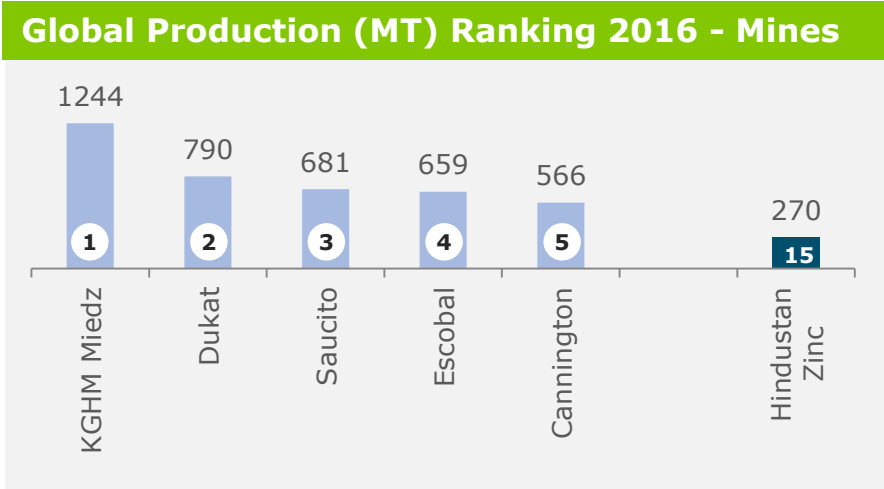
Silver – Fastest Growing Portfolio of Hindustan Zinc

Strong Management Focus to Drive Silver Portfolio

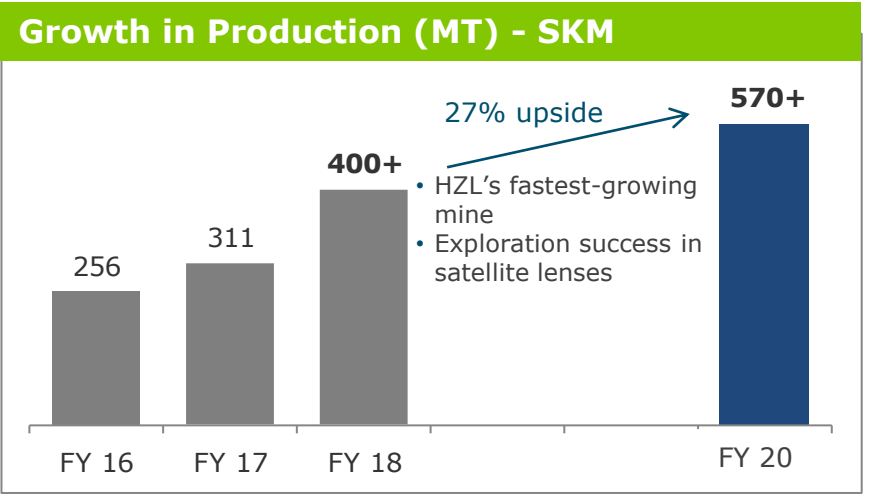
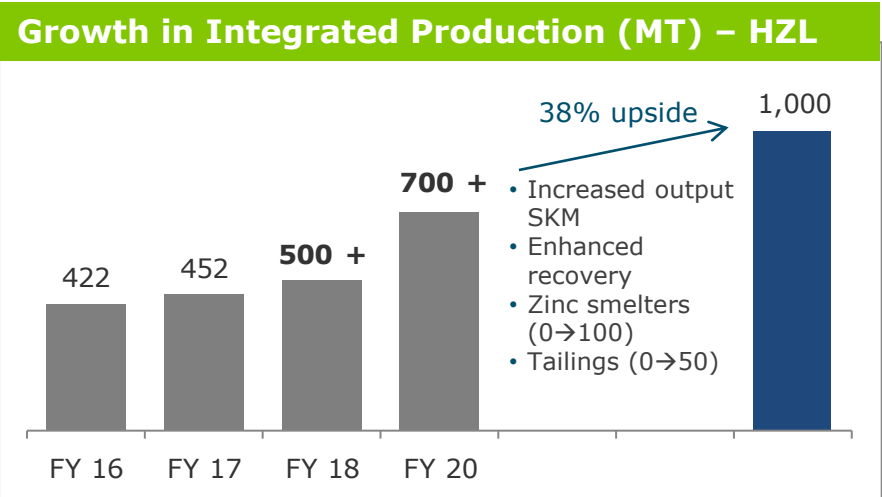




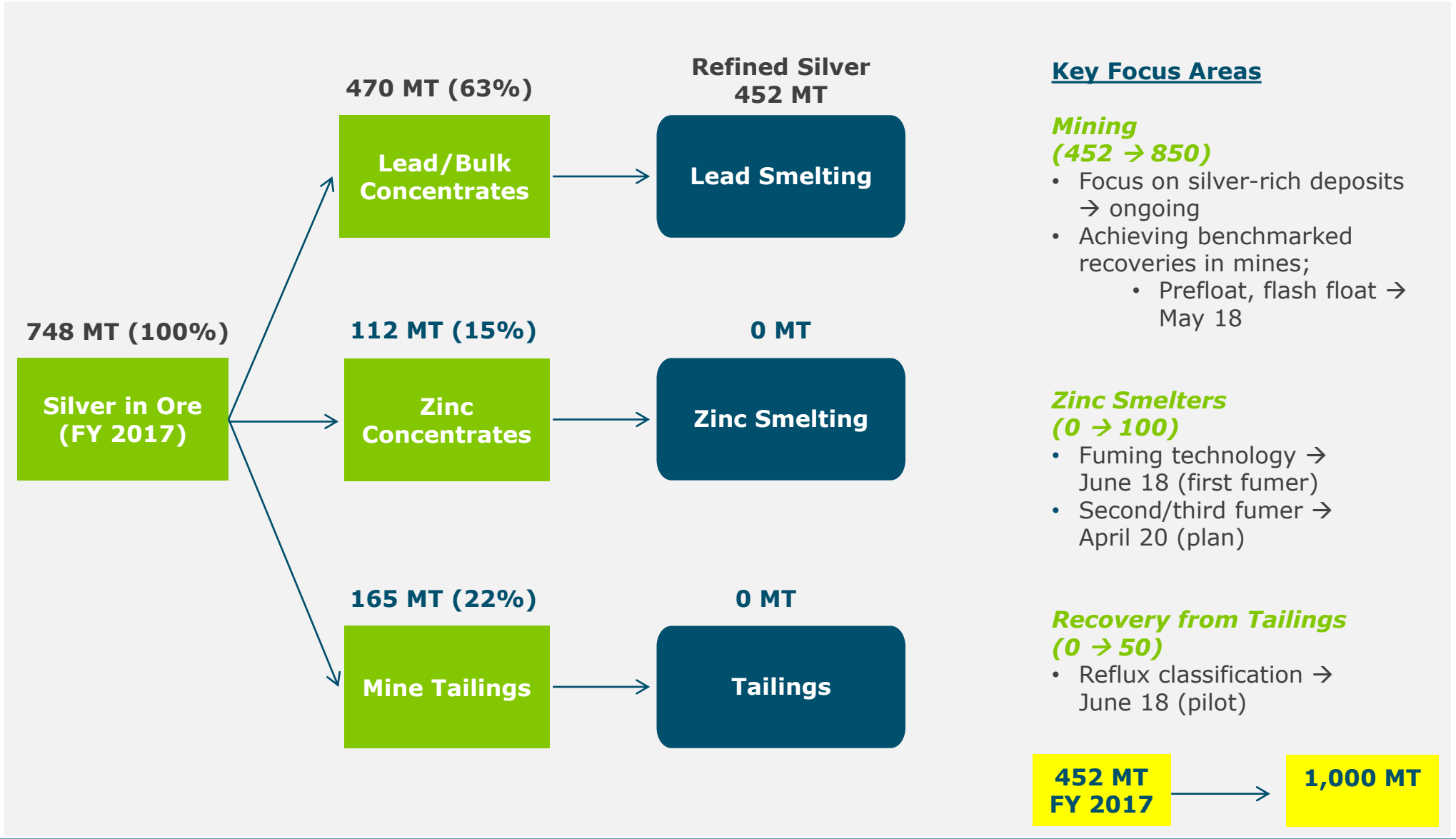
Source : Metals Focus and *Reuters



Source : Metals Focus



Hindustan Zinc Poised to be Among 'Top 5 Silver Companies'



Key Focus Areas

Mining (452 → 850)

- Focus on silver-rich deposits → ongoing
- Achieving benchmarked recoveries in mines;
 - Prefloat, flash float → May 18

Zinc Smelters (0 → 100)

- Fuming technology → June 18 (first fumer)
- Second/third fumer → April 20 (plan)

Recovery from Tailings (0 → 50)

- Reflux classification → June 18 (pilot)



06

Way Forward

Sunil Duggal,
CEO

**1.5 mtpa
capacity
under
planning**



Shaft and decline integration
Multi-level mining
Digital transformation
New prospects & tenements
R&R addition

**1.2 mtpa
capacity
by FY 2020**

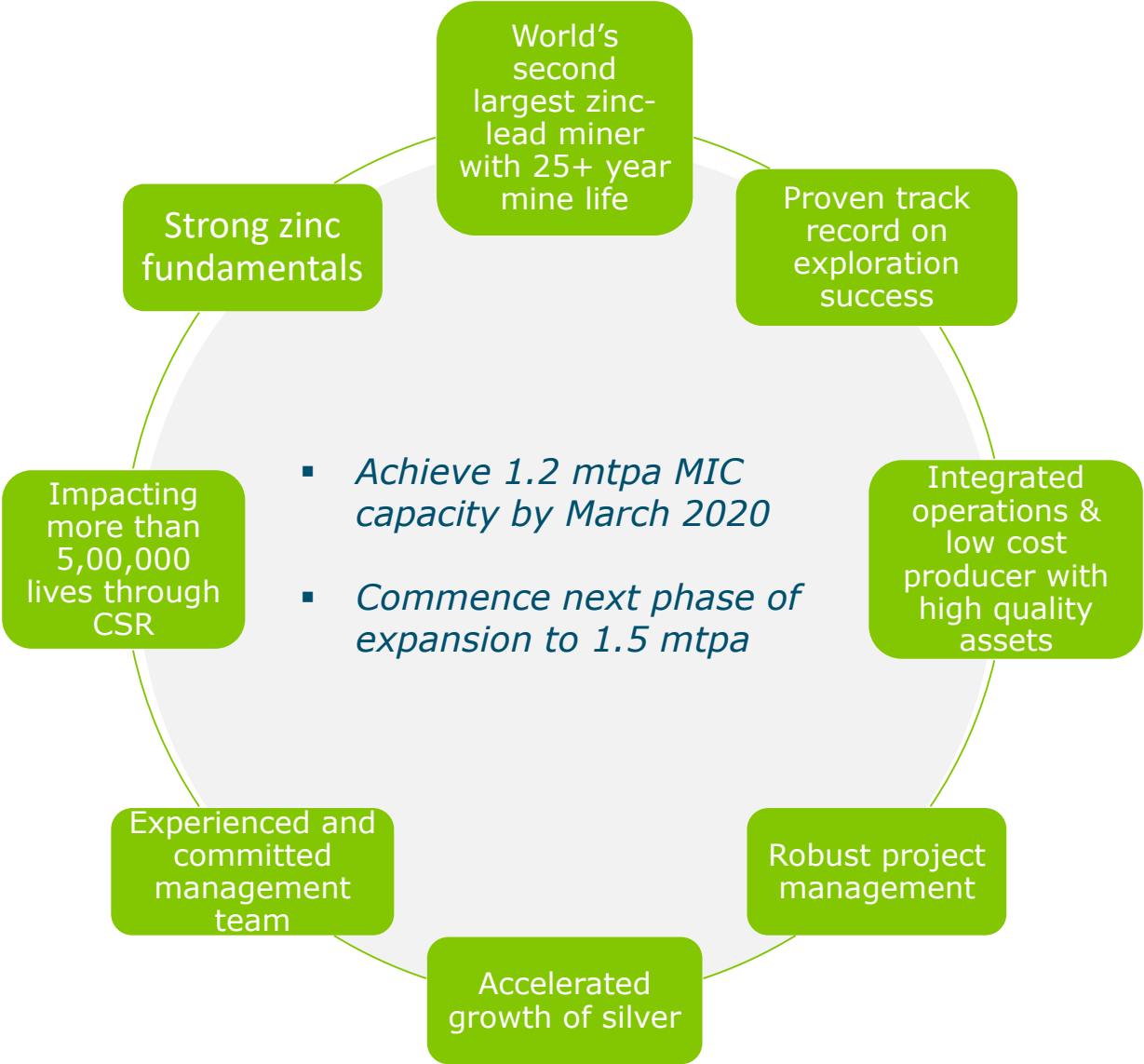


25+ years of mining R&R with potential of 1.2 mtpa
Transition to underground mining progressing smoothly
Shaft commissioning on track for Q3 FY 2019
Mill & smelter capacity to keep pace
Reduction in COP on shaft commissioning

**Outlook
FY 2018**



Mined metal: Higher than 907 kt of FY 2017
Refined zinc-lead: 950 kt; silver: 500 +MT
Marginally higher dollar COP (excluding royalty)
Project Capex of \$350-360 million (including ongoing mine expansions, fumer and smelter debottlenecking)





Thank You

Sample Financial Working & Sensitivity



INPUTS	S. No.	Particulars	UOM	Value
	1	Mined metal production	kt	1,200
	2	Metal production - Zinc	kt	979
	3	- Lead	kt	173
	4	Silver production	kt	750
	5	Realisation* - Zinc	\$/MT	2,950
	6	- Lead	\$/MT	2,350
	7	-Silver	\$/Troz	17.00
	8	Cost Of Production(COP) - Zinc	\$/MT	800
	9	- Lead	\$/MT	700
	10	- Silver	\$/Kg	20
	11	Royalty - Zinc		13.20
	12	- Lead	%	19.14
	13	- Silver		9.24
	14	Exploration and others		30
	15	Sustaining capex	\$ mn	80
SAMPLE FINANCIAL WORKING	16	Project capex		320
	17	Revenue - Zinc		2,889
	18	- Lead		406
	19	-Silver		410
	20	Total Revenue		3,705
	21	COP - Zinc		783
	22	- Lead		121
	23	-Silver		15
	24	Total COP		919
	25	Royalty - Zinc	\$ mn	381
	26	- Lead		78
	27	-Silver		38
	28	Total Royalty		497
	29	EBITDA Metal		2,258
	30	EBITDA Wind Power		25
	31	EBITDA Total		2,283
	30	Operational Free Cash**		1,916

*Includes conservative LME & Premium

**Excludes treasury income and working capital movement

Cost Sensitivity

Increase in LME

	EBITDA (\$ mn)	
Zinc (\$100/MT)	82	↑
Lead (\$100/MT)	17	↑
Silver (\$1/Toz)	20	↑

Every \$10/MT increase in COP

Zinc	9	↓
Lead	2	↓

Every 10kt increase in volume

Zinc	19	↑
Lead	13	↑

All the above are individual Sensitivities and assume that the other parameters in the sample calculations stay constant

Open Pit

Ultimate Pit Depth : 400 metres from surface

UG Mine

- **Planned Production Capacity** : 4.5 mtpa
- Fully mechanised trackless mining operation
- **Mining Method** : LHS with paste fill

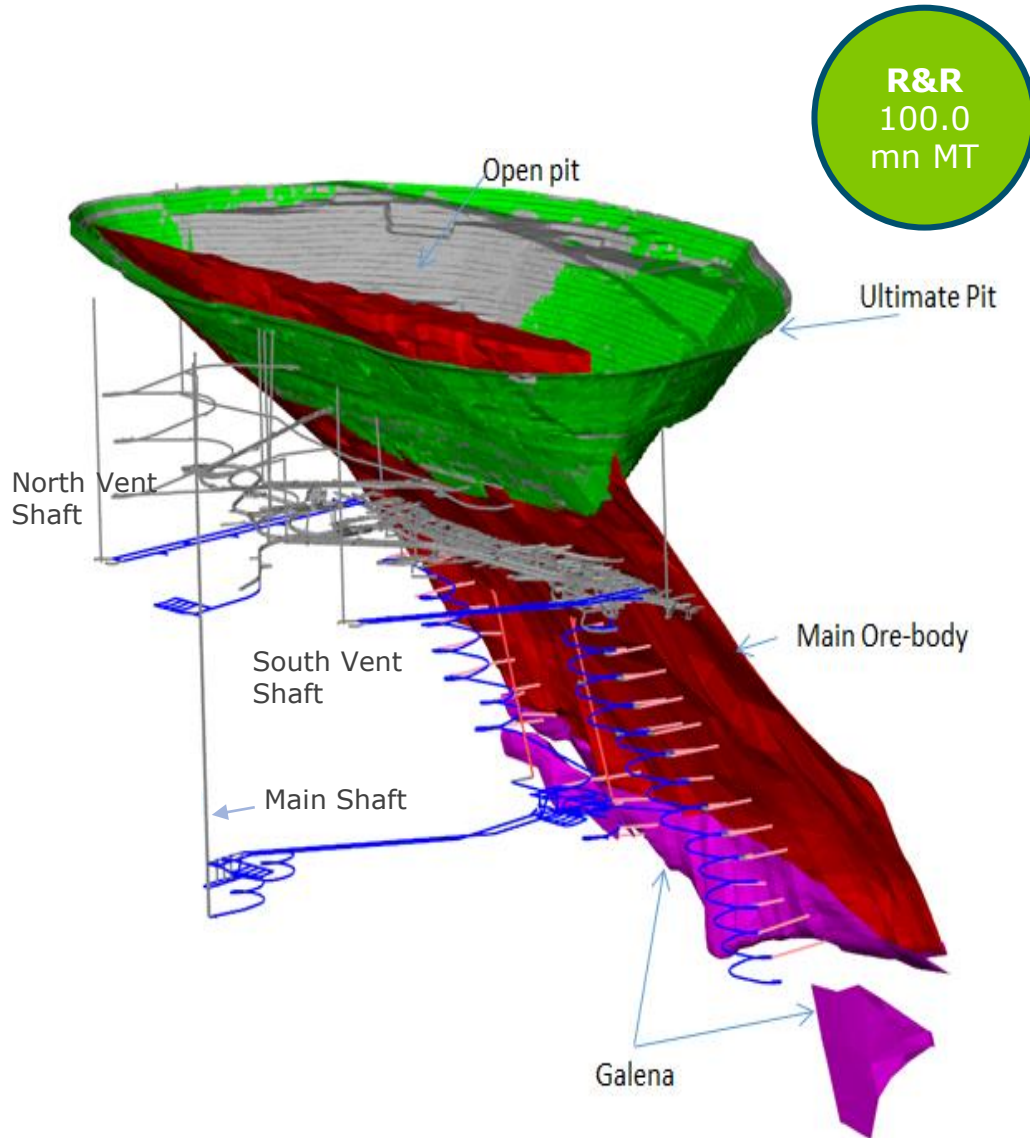
Major Activities

- Decline and ancillary development
- Production shaft and crusher system
- Raise boring
- Paste fill plant
- Surface & underground infrastructures

Shaft

- Main shaft 7.5 metres dia, capacity 3.75 mtpa, 955 metres deep
- 2 nos. ventilation shafts of 7.5 metres dia

Year	Milestone
2013	• North and South ramp started • Main shaft sinking started
2015	• Paste fill plant installed • North vent. shaft completed
2016	• Production from main block • Main shaft sinking completed • South vent. shaft completed
2017	• Shaft winder commissioning
2018	• Main shaft commissioning



UG Mine

- **Planned Production Capacity** : 6.0 mtpa
- **Mining Method** : Long hole stoping with paste filling
- Fully mechanised trackless mining operation

Major Activities

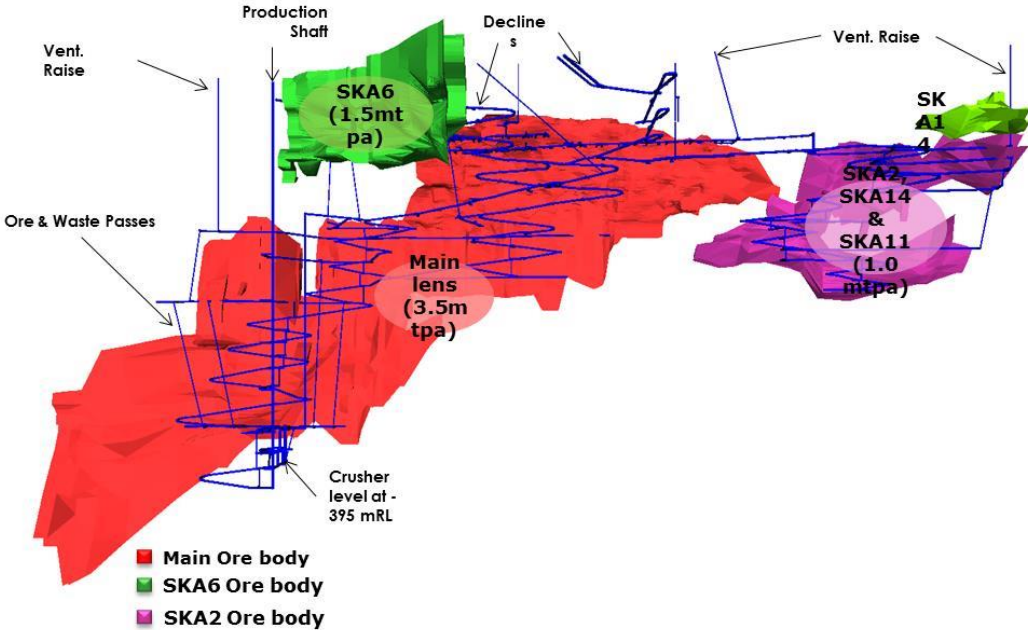
- Decline and ancillary development
- Production shaft and crusher system
- Raise boring
- Paste fill plant
- Beneficiation plant
- Surface and underground infrastructures

Shaft

- Main shaft 7.5 metres dia, capacity 3.75 mtpa, 1,050 metres deep

R&R
122.8
mn MT

Year	Milestone
2013	• Main shaft sinking started
2014	• South ramp commenced • Paste fill plant started
2015	• Paste fill plant commissioned • Mill DBN (2 to 2.8 mtpa) completed
2016	• Production from SKA6 • Main shaft sinking completed
2017	• Production from SKA14 • Mill plant (1.5 mtpa) commissioned
2018	• Shaft and winder commissioning • Mill (1.5 mtpa) commissioning

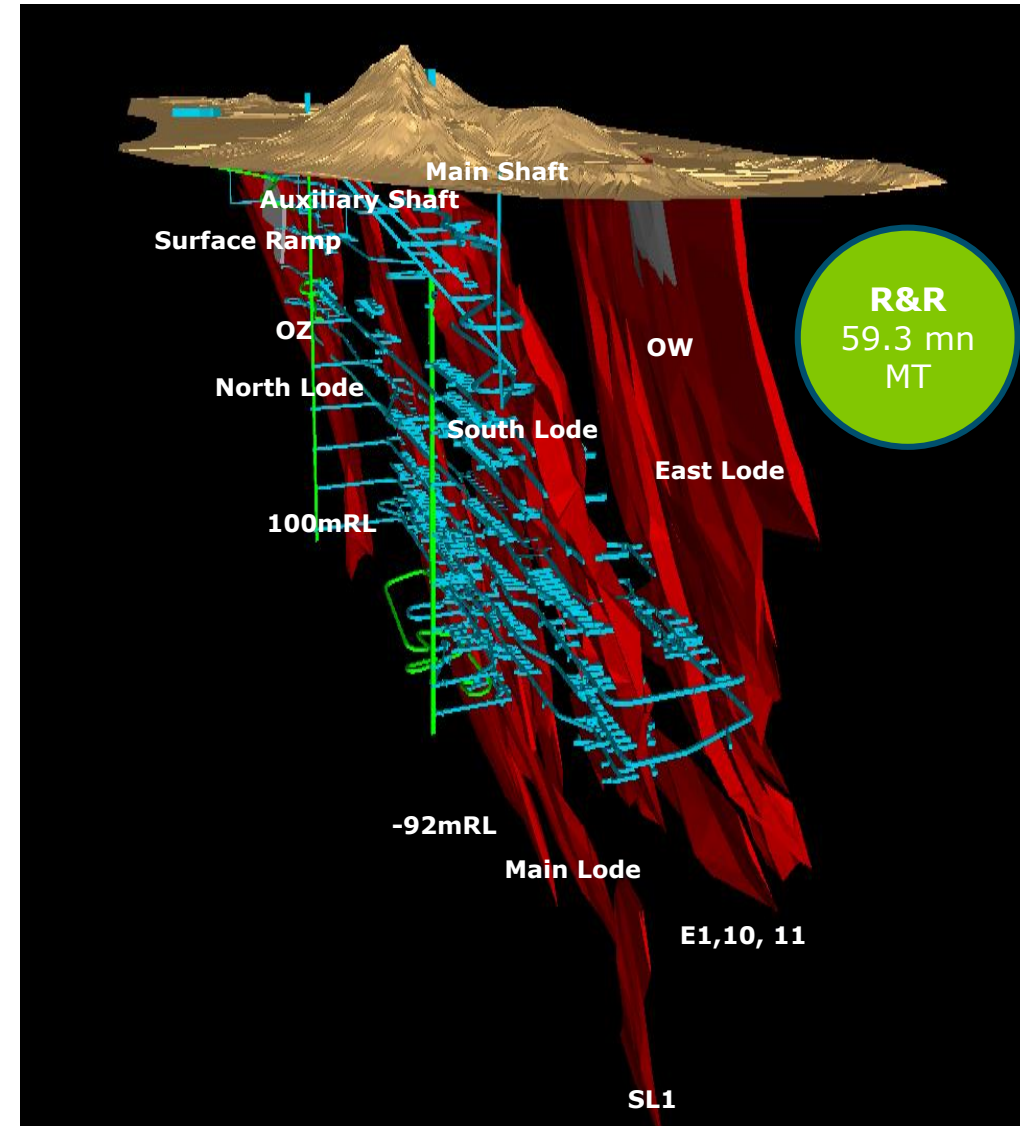


UG Mine

- **Access to Mine** : Through ramp and shaft
- **Method of Working** : Blast hole stoping with hydraulic filling
- **Ore Hauling** : Through ramp and shaft

Major Activities

- Ramp up to 1.5 mtpa ore production capacity
- Decline and ancillary development
- Raise boring
- Hydro fill plant
- Debottlenecking of beneficiation capacity
- Surface and underground infrastructures – new and upgradation



UG Mine

- Cluster of 4 mines having independent infrastructure
- **Access to Mine** : Through ramp and shaft
- **Access to Mine** : Through ramp and shaft
- **Method of Working** : Blast hole open stoping
- **Ore hauling** : Through ramp and shaft

Major Activities

Ramp up to 4.5 mtpa ore production capacity

- Declines and ancillary development
- Crusher system
- Raise boring
- Beneficiation plant
- Surface and underground infrastructures
- Introduction of mechanised mining
- New declines under development for trackless mining

Year	Milestone
2015	• Mochia, Balaria and North Baroi mine dev commenced
2016	• Surface infrastructure at North Baroi
2017	• North Baroi decline completion • Mill DBN (1.2 to 2.7 mtpa) commissioning
2018	• Mochia decline completion • Mill (2 mtpa) commissioning
2019	• Balaria decline completion • Back fill plant commissioning



UG Mine

- Shallow Ore Body
- **Access to Mine** : Through ramp
- **Mining Method** : Long hole open stoping with filling
- **Ore Hauling** : By 50 T LPDT through ramp
- **Ore Treatment** : By 50 T LPDT through ramp

Major Activities

- Decline and ancillary development
- Raise boring
- Cement Rock fill plant
- Surface and underground infrastructures

